OpenPrivateer Player Manual

Public Alpha

OpenPrivateer is a modern reimplementation of *Wing Commander: Privateer*. It runs the game natively on modern systems while using the original game data.

OpenPrivateer does not include the original Origin/EA game assets. You need your own copy of *Privateer* and, if you want to play it, *Righteous Fire*.

This is a player manual. The reverse-engineering notes and file-format documentation in the source tree go much deeper.

1. Quick Start

1. Download and extract OpenPrivateer.
2. Start `openprivateer.exe` on Windows, or the corresponding executable on Linux or macOS.
3. The launcher opens. Select **Privateer** or **Righteous Fire** at the top.
4. Open **Game Data** and provide the original game files:
   - Privateer: `PRIV.TRE` and `INITCFG.PAK`
   - Righteous Fire: `RF.TRE` and `INITRF.PAK`
5. Use **Import to Data Folder...** to copy files from another folder, or **Add Folder...** to use files where they already are.
6. Click **PLAY**, then **NEW**.

The launcher shows the exact files it found and the folder each one comes from. You can also put the files next to the OpenPrivateer program; the launcher searches the Program Folder automatically.

The INIT files contain the original new-pilot starting state. If the matching INIT file is missing, **NEW** is disabled.

You do not need *Righteous Fire* to play the original campaign.

SDL3 on Linux and macOS

The Linux and macOS archives include the SDL3 library (`libSDL3.so*` or `libSDL3*.dylib`) next to the `openprivateer` program, and the program uses it automatically. To use your system's SDL3 instead (3.2.16 or newer), delete those `libSDL3` files from the program folder.

GOG installation

The GOG release includes both *Privateer* and *Righteous Fire*. In its installation folder, `INITCFG.PAK` and `INITRF.PAK` are ordinary files, and `PRIV.TRE` and `RF.TRE` are inside `GAME.GOG`, a disc image. The launcher extracts them for you:

1. Open **Game Data**.

2. If the launcher found your GOG installation, it shows “Found a GOG install in …” with an **Extract** button. Click it.
3. Otherwise click **Extract from GOG…** and select `GAME.GOG` in your GOG installation folder. **Import to Data Folder…** on that folder does the same.

`PRIV.TRE` and `RF.TRE` are copied out of the image, and `INITCFG.PAK` and `INITRF.PAK` from beside it, into the Data Folder. The progress bar shows how far it is; **Cancel** stops it and leaves no partial file behind. Each copy is checked against the image before it is used. If a different `PRIV.TRE` or `RF.TRE` is already in the Data Folder, the launcher asks before replacing it. If the image lacks one of the files, the others are still copied and the launcher says what is missing. The GOG installation itself is never changed.

Extraction runs on your own computer, from your own copy of the game. Nothing is downloaded, uploaded or shared, and OpenPrivateer never ships game files.

The launcher looks for GOG installations in the usual folders: on Windows `GOG Galaxy\Games\Wing Commander Privateer` and `GOG Games\Wing Commander Privateer` on every local drive, and the default GOG Galaxy folder under Program Files; on Linux and macOS `GOG Games/Wing Commander Privateer` in your home folder. Anywhere else, use **Extract from GOG…** and select the file: any location works. On macOS, if your GOG copy is an application bundle, `GAME.GOG` may be inside it; in Finder, right-click the bundle and choose **Show Package Contents** to find it.

2. First Start and the Launcher

The launcher is OpenPrivateer’s startup screen. It selects the game, locates the original game data and provides the main startup settings before the engine loads the game.

At the top, click the **Privateer** or **Righteous Fire** picture to choose the game. Your own `wallpaper/` artwork is used when present (section 19). On the left, the launcher shows the selected game’s TRE and INIT file, their status and the exact path being used. **PLAY** stays disabled until the selected game’s TRE file is found.

When a newer OpenPrivateer build is available, a line under the build number says **Update available** with the build’s name and a **Download** button. Download opens the newest release in your web browser; nothing is downloaded or installed automatically (section 23).

The launcher sections are:

- **Experience**: choose Classic, Enhanced or Modern (section 4).
- **Multiplayer**: enter a server to join and your callsign, and check the server before joining (section 22).
- **Game Data**: manage where OpenPrivateer finds the four original game files.
- **Mods**: switch installed mods on and off, set their load order, choose the start mod and see what the last start did with each mod (section 20, “Managing mods in the launcher”).
- **Music**: choose the music backend and an optional SoundFont (section 17).
- **Original Developer Switches**: set the original `-K`, `-B` and `-D` startup switches (section 24).
- **Files and Diagnostics**: open user-data folders, move the user-data location, reset settings, copy diagnostics and control update checks.

Untick **Show this screen at startup** to go straight to the game next time. Hold **Shift** while starting OpenPrivateer to bring the launcher back. The launcher is always shown when required game data is missing.

Game Data

OpenPrivateer searches the Game Data rows from top to bottom and uses the first copy of each required file it finds. The table shows all four files and marks the copy currently selected as **Used**.

The **Type** column describes each search location:

- **Custom**: a folder added by the player. Its files are used in place.
- **Data Folder**: the `data/` folder where the launcher stores imported or copied game files.
- **User Home**: the current user-data folder.
- **Program Folder**: the folder containing the OpenPrivateer program. In portable mode this is also the user-data folder.

Command-line and environment search locations are identified by their corresponding source instead.

Other table states help explain duplicate files. **differs** means a different copy exists further down the search path but is not used. **same** means the later copy is identical. **case!** reports a filename with the wrong letter case on systems where that matters.

Folders added as **Custom** can be moved up or down or removed. A path can also be entered directly and added with **Add**. **Rescan** checks the search locations again.

The four file actions are different:

- **Add Folder…** adds a folder to the search path and leaves its files where they are.
- **Import to Data Folder…** selects a folder and copies recognized game files from it into the Data Folder (from a GOG installation folder, it extracts them from `GAME.GOG`).
- **Copy Found Files to Data Folder** copies the game files currently being used into the Data Folder.
- **Extract from GOG…** selects a GOG `GAME.GOG` and extracts the game files from it into the Data Folder (section 1, “GOG installation”).

When OpenPrivateer recognizes a likely existing installation, the launcher can also offer to use its files in place, copy the files it finds, or, for a GOG installation, extract them.

Where your files live

Settings (`openprivateer.ini`), savegames, logs and optional folders live together in OpenPrivateer’s user-data folder.

By default this is:

System	Folder
Windows	`%APPDATA%\OpenPrivateer`
Linux	`~/.local/share/OpenPrivateer`
macOS	`~/Library/Application Support/OpenPrivateer`

A portable installation instead keeps user data in the Program Folder. An existing `openprivateer.ini` next to the program keeps older portable installations working. An empty `portable.txt` can also be placed there to request portable mode.

Open **Files and Diagnostics** to see the folder currently in use. The section has direct buttons for the user-data folder, savegames, logs, music, wallpaper and content.

Change Data Folder... selects another location for user data. When switching folders, OpenPrivateer can either use the destination as it is or copy the current settings, saves, content, music, wallpaper and copied game files there before switching. Nothing is deleted from the old location. Depending on the current mode, the launcher also offers **Move to User Folder...** or **Use User Folder...**.

If the location was fixed with `--home` or `OPENPRIV_HOME`, it cannot be changed from the launcher for that run.

Reset Settings... renames `openprivateer.ini` to `openprivateer.ini.bak` and restarts OpenPrivateer. Savegames are not touched.

OpenPrivateer creates its optional folders and README files in the user-data folder automatically. The README files are rewritten on startup so their instructions stay in sync with the installed engine.

Folder	Purpose
`savegame/`	Original and OpenPrivateer savegames
`music/`	Optional replacement music
`wallpaper/`	Optional game-selection artwork
`content/`	Content overrides, translations, wingman names, the
	experimental `content/mods/universe/` map extension,
	the experimental `content/mods/weapons/` weapon mods,
	`content/mods/items/` Ship Dealer mods,
	`content/mods/init/` start mods,
	`content/mods/missions/` mission mods,
	`content/mods/ships/` ship mods,
	`content/mods/interiors/` base interior mods and
	`content/mods/packs/` mod packs (see section 20), and
	`content/hud/hud_style.json`, your own style for the
	cockpitless HUD (see section 14)
`mods.json`	Optional: the mods' load order and the disabled mods
`alpha/`	Optional material from the August 1993 Privateer alpha
`data/`	Game files imported or copied by the launcher
`logs/`	Session logs

You can ignore all of these folders when starting out.

3. Privateer and Righteous Fire

OpenPrivateer supports both the original *Privateer* campaign and *Righteous Fire*.

Choose the game by clicking its picture at the top of the launcher. With the launcher turned off and both `PRIV.TRE` and `RF.TRE` found, a startup selector lets you choose instead. The selection is remembered. You can also change the game later from the settings/debug overlay. Changing games while playing relaunches OpenPrivateer because the two modes use different data archives.

Importing a Privateer pilot into Righteous Fire

The original Righteous Fire import flow is supported.

If Righteous Fire is selected and the `savegame/` folder contains Privateer `.SAV` files, starting **NEW** can ask whether you want to load an old Privateer ship.

The import carries over the pilot, ship, credits, reputation, kills, guild memberships and missions-to-date state according to the original import behavior. Privateer cargo, side missions and plot progress do not become Righteous Fire campaign state.

Righteous Fire equipment in Privateer

With `RF.TRE` in the game data folder, a Privateer game can sell Righteous Fire's equipment at the Ship Dealer: turn on **All Equipment** in the debug overlay (directly above Open Universe). It is off by default, works in Enhanced and Modern, and takes effect at once. See section 11 for what it adds.

4. Experience Modes

OpenPrivateer has three experience settings: **Classic**, **Enhanced** and **Modern**. Enhanced and Modern currently use the same gameplay behavior; Modern is reserved for future changes.

Classic

Classic aims to reproduce the original gameplay, mechanics and feel. It is not a DOS emulator and does not deliberately reproduce every old hardware limitation or arithmetic accident. OpenPrivateer-only gameplay features such as multiple-ship ownership are inactive.

Enhanced and Modern

Enhanced and Modern keep the original game underneath but add OpenPrivateer conveniences and optional features.

Flight

- **Escape lands at a nearby base:** `Esc` in flight lands you at the nearest base, but only while you are inside that base's landing zone (5000 units). With no base that near it does nothing. In Classic, `Esc` does nothing in flight.

- **Hail to land:** flying close to a base or planet does not land you by itself. Ask for a landing instead: inside the base's landing zone (5000 units) press `Esc`, or pick the base on the Communications MFD and send "Requesting immediate automatic docking", which lands you at once. At a carrier (section 20) that line appears in the menu only while you are within 2000 units of its deck, at the top of the list, and goes again if you move farther away, so the other lines move down or up by one. The base does not announce its landing zone. The flight music keeps playing near the base; the landing music plays during the landing movie. To land automatically as in Classic (within 3500 units, no request needed, with "Entering automatic landing zone" and the landing music on entering the zone), untick **Hail To Land** in the GAME section of the settings overlay. Classic always lands automatically.
- **Bases by name on the Communications MFD:** the list of contacts names each base and planet as the Navigation Map does ("Jolson" rather than "Agricultural Base"). A name too wide for the MFD ends in "…". Ships keep their usual names, and comm messages still start with the base's usual comm name. Classic lists bases as the original does.
- **Cockpit hide:** `.` hides the cockpit background art while the HUD stays visible, in every cockpit view including the turret views (`F5` / `F6`). The setting applies to all views at once and lasts until you press `.` again or launch next. It is unbound in Classic.
- **Red crosshair on the lead:** in the front view the crosshair of every cockpit turns red, the red of the box around an enemy target, while it is on a gun's lead marker (the ITTS markers, switched with `I` on scanners that have them) and for half a second after, so even a brief pass shows. A shot then is aimed where the target will be. Classic keeps the crosshair's own colour.
- **ITTS plaque:** `I` shows "ITTS Engaged" or "ITTS Disabled" for 3 seconds at the top of the screen, where F7's "Missile Camera On" plaque appears, following the **Plaque Position** setting (Off hides it). Classic shows nothing.
- **Cockpitless HUD:** `,` (comma) switches between the cockpit and a HUD drawn as thin lines over open space, on every ship (see "The cockpitless HUD" in section 14). Each flight starts in the cockpit; it is unbound in Classic.
- **Rolled sprites keep their shape:** a planet stays round at any roll. Classic keeps the original oval distortion.
- **The keyboard steers alongside your input device:** the keyboard adds to the device selected on the Options screen, so you can, for example, roll with the keypad keys while turning with the mouse or stick. The mouse flies the ship only with Mouse selected and a joystick or gamepad only with Joystick selected. In Classic only the device selected on the Options screen steers. Each device's own roll works the same in every mode: the mouse's right button and the original joystick's button 2 switch from turning to rolling, while a gamepad's right stick or a joystick axis assigned to Roll rolls while you turn.
- **Mouse flight is easier to control:** one even dead zone, the same sensitivity on both axes, and pushing the mouse up pitches the nose up. A second crosshair follows the pointer so you can see where you are steering, and **Mouse Sensitivity** applies. Classic keeps the original mouse flight: no visible pointer, and mouse up pitches the nose down.
- **Weapon state is remembered across a launch:** the gun on/off combination, last selected ordnance type, ITTS and shield power level (`S` / `Shift+S`) are restored for as long as the game is running. ITTS only comes back on a ship whose scanner has ITTS. Loading a save or starting a new game begins from the defaults. Classic launches with all guns off, ITTS off and shields at full power.
- **Gun keys 1-4 address hardpoints:** each key flips one forward hardpoint. In Classic they flip a weapon-type group, so four identical guns all flip together on `1`.

- **Autopilot switches the tractor beam off.** In Classic it stays on.
- **Autonomous turrets:** a turret with no manual input slews toward the nearest hostile ship, leads it and fires. In Classic turrets are aimed by hand only.
  - Each turret has its own energy pool, so it cannot starve the forward guns. Classic uses one shared pool.
  - NPC turrets get a gunner too. In Classic they are fixed to the hull. **AI Turret Lead** in the debug overlay controls whether they lead.
- **Turret targeting uses the turret's aim:** `T` and auto-acquire in a turret view use the turret's own aim. Classic uses the ship's nose.
- **Other ships' missile and torpedo launches are audible** within 5000 units. In Classic they are silent.
- **AI Missile Locks** in the debug overlay (off by default): while it is on, a computer-flown ship's Heat Seeker only homes when fired from behind its target, and its Dumb Fire flies straight. Off, as in the original, their missiles home from any angle. It has no effect in Classic or in multiplayer.
- **Skipping a cutscene consumes the input:** the key or click that skips it does not also fire your guns. In Classic it also reaches the game, as in the original.
- **More cutscenes can be skipped:** the autopilot flyby and death cutscene can be skipped by any key or click. They cannot in Classic. The jump charge-up cannot be skipped in any mode.

Navigation

- **The Navigation Map opens while docked:** `N` opens it from any docked screen except the Mission Computer, where `N` types into the save name. Classic allows it only in flight.
- **Mission progress in flight:** when you complete a mission objective, the cockpit message line says so, for example **GO TO DELTA, NAV 2: COMPLETE**. See section 16. Classic shows nothing.
- **Right-click zooms out:** Quadrant View to quadrant overview, then quadrant overview to sectors overview. This needs **Open Universe**.
- **Arrow keys walk between quadrants** in Quadrant View. In the other map views they move the pointer. The numeric keypad and the D-pad move the pointer in every view.
- `Home` **returns to the current system** in Quadrant View.
- `Shift` + `N` **goes backwards** through the nav points, the same way `N` (NEXT) goes forwards, wrapping from the first back to the last. In Quadrant View it steps back through the quadrants, and on the mission page back through your missions.
- `Shift` **+click a system** to show only its jump lines and list its destinations. A plain click shows every line again.
- **Hold** `Ctrl` to grey out systems your remaining jump fuel cannot reach, and `Ctrl` **+click** a system to see how many jumps away it is.
- **Hold** `Shift` to colour systems by faction.
- **Open Universe is on by default:** mod systems, jumps, bases and zones can become reachable. It is inactive in Classic. Built-in content currently adds Gemini in Humboldt, with the planet Gemini, the Heliopolis Array and Novaya Riga stations, two asteroid fields, random encounters and jumps to Troy, Junction and Aquila. **Open Universe** and **Built-in Content** in the debug overlay control this content. See section 20.

Bases and screens

- **Escape backs out one level** on base screens. It does nothing in Classic.

- **Mission Computer from Repair & Upgrade** is allowed. Classic blocks it.
- **Mission Computer REJCT button:** the alternate keypad can remove a mission from your list, docked or in flight. In flight the mission's own ships leave at once; your other missions keep their progress. Classic has only the original keypad.
- **No unreachable mission offers:** without a jump drive, the First Sector Mission Machine and both guild boards offer only missions inside the current system. A board with nothing local to offer says so; a guild computer adds that a jump drive would open up missions in other systems. The Merchants' Guild, whose work is almost all cargo runs and bounties elsewhere, is often empty until you fit a jump drive. As on every board, the offers you see last until you next launch: missions you accept leave the board, and new offers appear only after your next landing. Classic offers missions anywhere, as the original does.
- **Harder missions pay more:** the pay offered rises with how tough the encounter will be, which grows with your credits, your missions flown and your ship (and is higher at the Mercenaries' Guild). A new pilot sees the same pay as in Classic; it is never less. Classic pays exactly what the original does.
- **Side missions survive unrelated landings:** they are consumed only when they actually complete and pay. Classic drops every carried side mission on any landing.
- **Notification center:** press **M** on a Launch Pad to read the results of your missions. It also opens by itself after a landing that completed or failed a mission. See section 16.
- **Cargo Manifest last page:** it lists its items instead of the original's empty last page when the hold has exactly 8, 16 or 24 item types.
- **Bar brush-offs are local and temporary:** only that bartender or patron is affected, until you leave. In Classic one brush-off silences every bar for the rest of the career.
- **Entrance slides are quicker.** Classic uses the original 10 steps.
- **Clearer duplicate-purchase message:** buying a single-instance item you already have says **ALREADY OWNED**. Classic reuses the original **NO ROOM ON SHIP** message.
- **Higher ordnance capacity:** missile and torpedo capacity is 20 per matching launcher. Classic uses the original 10 per launcher.
- **Faster hardpoint-purchase cancel:** right-click cancels the Ship Dealer hardpoint picker. Classic keeps only the on-screen **Never mind** region.

Enhanced and Modern also permit multiple ships and hired wingmen when their settings are enabled. Use **Classic** for direct comparison with DOS Privateer, and **Enhanced** or **Modern** for OpenPrivateer's extra features.

The Experience setting can be changed while playing. In multiplayer it is fixed by the server.

5. Settings and Debug Overlay

During the Alpha, some settings still live in the debug/settings overlay rather than a finished front end.

Press:

```
~
```

This is normally the key to the left of `1` on an English keyboard. Its label depends on your keyboard layout.

Choosing another overlay key

If your keyboard has no `~` key, or it does not reach the game (some on-screen and emulated keyboards), you can choose another key. With OpenPrivateer closed, add a line like this to `openprivateer.ini`:

```
DebugOverlayKey=F12
```

That key then opens and closes the overlay instead of `~`, everywhere `~` would: in flight, on bases, in menus, the launcher, and while typing in one of the overlay's own text fields. The overlay's hint line names the key in use.

The value is a key name, optionally preceded by `Ctrl+`, `Alt+` and `Shift+`. Upper and lower case do not matter. Examples:

Value	Key
`F12`	F12
`Insert`	Insert
`\`	the backslash key
`Ctrl+F1`	F1 while Ctrl is held

The names are SDL's key names, the same ones the Controls screen writes to `openprivateer.ini` for keys you remap. Most keys are named as they are labelled (`F1`, `Home`, `Keypad 5`); a few punctuation keys are named by their character, such as `\` for backslash and `` ` `` for the key left of `1`.

With modifiers in the value, exactly those must be held: `Ctrl+F1` does not react to F1 alone or to Ctrl+Shift+F1. If the key also has a game action (for example `\`), it opens the overlay and the game action no longer reacts to it; the session log names that action at startup.

If the value is not a key name, `~` stays the overlay key and the session log says so at startup. An empty value also keeps `~`. Delete the line to go back to `~` for good. OpenPrivateer keeps the line as you wrote it and never adds it on its own.

The overlay contains settings for areas such as Experience mode, Privateer / Righteous Fire mode, audio, SoundFont, rendering, subtitles, Mission Computer style, controllers, Righteous Fire equipment in Privateer (All Equipment), multiple ships, hired wingmen, save file naming (Preserve Savegames, see section 7), the mod switches (Weapon, Item, Mission, Ship, Interior and Init Mods, Open Universe and Built-in Content, see section 20), Hull Atlas (section 18), the cockpit message line's position (Plaque Position, section 14) and update checking.

Some settings are disabled in Classic because they would change original gameplay.

6. Controls

OpenPrivateer supports keyboard, mouse, gamepads, joysticks and HOTAS-style controllers.

Open **Controls...** from the settings interface to inspect or remap controls. Controller bindings are stored per device. The same screen can also be reached by pressing `K` on the Options screen, opened in flight or on a base with `Alt+O`.

Assigning a controller

Selecting a controller and assigning it are separate steps:

1. Open **Controls...**, then **Remap Game Controls**.
2. Choose the physical controller from the controller drop-down at the top.
3. Click **Assign Controller**.

The drop-down only selects which connected device the Controller column shows and edits. **Assign Controller** makes that device active in OpenPrivateer. With several controllers connected, check the drop-down before changing a binding.

Assigning buttons, hats and axes

After assigning the controller:

1. Find the game action you want to change.
2. Click one of its slots in the **Controller** column.
3. Press the desired button, move the hat, or move the desired axis.

For example, to assign a joystick X axis to Yaw: select and assign the joystick, click the Controller slot on the **Yaw** axis row, then move the joystick's X axis.

Right-click a binding slot to clear it. `Esc` cancels capture without changing the slot. If the input you press is already used by another action that works at the same time (for example two flight actions), it is moved to the new action and the status line says where it was removed from. Actions that never work at the same time can share an input: a key can mean one thing in flight and another on a base screen or in the Mission Computer. Global keys such as fullscreen and quit work everywhere, so they share with nothing.

To put several actions on one input, tick **Allow one input on several actions** on the Settings page. A new binding then keeps the other actions on that input, and the status line says which ones ("Also used by: ..."); inputs used by more than one action show in orange on the remap page. When you press a shared input, the action that can act right now gets it: with Jump and Autopilot on one button, the button jumps inside a jump point and engages the autopilot anywhere else. Actions that act while held (steering, guns, afterburner) all act together. Global keys and whole controller axes cannot be shared. Turning the option off keeps the shared bindings you already made; new bindings then take their input over again. A button or key you are still holding when it starts the autopilot, a jump, a landing or the self-destruct does nothing more until you let go of it.

Clear Controls clears the selected controller's bindings. **Remove** makes it inactive; it stays in the list and can be assigned again later.

OpenPrivateer can bind controller buttons, hats, axis directions and full analog axes. On Pitch, Yaw, Roll and Throttle, the round button next to the controller axis reverses it (filled = reversed).

Calibrating an axis or throttle

A full controller axis must be assigned before **Calibrate** becomes available. The complete order is therefore:

1. Select the controller in the drop-down.
2. Click **Assign Controller**.
3. Click the Controller slot on Pitch, Yaw, Roll or Throttle.
4. Move the physical axis to assign it.
5. Click **Calibrate** on that row.

The dialog shows the live raw value. Confirm with `Enter`, **Confirm**, or a tap of any button on that controller. To cancel, hold any button on that controller for about 1.5 seconds (a bar fills while you hold it), or press `Esc` or **Cancel**. Cancelling keeps the previous calibration.

- **Pitch, Yaw, Roll:** move through the full travel in both directions, let the control rest at centre, then confirm. OpenPrivateer records minimum, centre and maximum.
- **Throttle:** confirm three points: **zero**, **full military power** and the point where **afterburner** should begin. Choose **No afterburner zone** if the lever has no afterburner detent. The order establishes the lever direction, so a calibrated throttle normally does not need the reverse control.

Throttle axis modes

The Throttle row has a **Mode** button:

- **Auto** (default): Absolute for a joystick/throttle axis, Relative for a gamepad stick.
- **Absolute:** lever position directly sets speed. Moving past full military power into a calibrated afterburner zone engages afterburner.
- **Relative:** deflection from centre raises or lowers speed, like the original joystick throttle. On a calibrated throttle the calibrated **zero** point is the centre: moving towards full military power raises speed, moving the other way lowers it.

In Absolute mode the lever only takes over after you move it. Starting a flight does not immediately force the ship to the lever's physical position. Keyboard and mouse throttle controls still work; moving the lever again gives it control.

Calibration

OpenPrivateer also supports general joystick calibration. A raw joystick needs at least two axes and one button to qualify as a flight controller. SDL-recognized gamepads are accepted directly.

If several devices are attached, the calibration screen calibrates the **Primary Flight Controller** (see Controller settings). With **Automatic**, devices that cannot actually fly the ship are skipped.

The first time the calibration screen appears on its own (Options set to Joystick and the controller never calibrated), the Controls settings open on top of it. Pick your Primary Flight Controller there and press

Done: the calibration screen is waiting underneath and calibrates the controller you chose. While the settings are open, the calibration screen ignores the controller.

On the calibration screen, tap a button to confirm each step. Hold a button for about 1.5 seconds to cancel; a bar fills while you hold it. `Esc` also cancels. Cancelling works at any step in every experience mode and keeps the previous calibration.

Controller settings

- **Dead Zone:** in Enhanced and Modern a round dead zone around the stick centre, adjustable from 3 to 30 percent (default 15), so a diagonal push is not cut off on one axis. Classic uses the original 30 percent dead zone on each axis separately.
- **Response Curve:** Linear or Smooth in Enhanced and Modern. Smooth gives finer control near the centre and full rate at full throw.
- **Mouse Sensitivity:** for mouse flight in Enhanced and Modern.
- **Input Backend** on Windows: WinMM or SDL DirectInput.
- **Primary Flight Controller:** which controller is "the joystick". It flies with the original stick scheme (trigger, button 2 roll, double-tap afterburner), moves the menu pointer, is the one the whole-joystick calibration screen calibrates, and in Classic is the Joystick input device. **Automatic** (the default) picks the first connected controller that can fly. The list updates when you plug a controller in or out; two identical controllers show as "(1)" and "(2)". The choice applies immediately. It does not change any button or axis binding: those stay with the controller they were assigned on, set with **Assign Controller**. If the chosen controller is not connected, the game uses Automatic's choice for now, keeps your choice for next time, and the page says which controller it is using. A newly chosen controller that has never been calibrated is not calibrated automatically; use the calibration screen for it.

Default gamepad layout

Input	Default action
Left stick Y	Pitch
Left stick X	Yaw
Right stick X	Roll
Right stick Y	Throttle
Right trigger	Fire guns
Right shoulder	Fire ordnance
Left trigger	Afterburner
Left shoulder	Cycle target
South button	Target lock
East button	Cycle ordnance
West button	Cycle guns
North button	Toggle ITTS

Input	Default action
Start	Pause
Back	Navigation Map
D-pad up	Front view
D-pad down	Rear view
D-pad left	Left view
D-pad right	Right view

These are normal remappable bindings. A joystick or HOTAS that SDL does not recognize as a gamepad gets the same four views on its first hat switch. If you have put something else on a D-pad direction, your binding stays and that view default is left off. The views are flight-only; on the menu screens the D-pad (and the first hat) moves the pointer in Enhanced and Modern.

Joystick pointer

A calibrated joystick can move the pointer on base and menu screens, matching a feature of the original game. The joystick trigger acts as the click. SDL gamepads use the A/South button.

Keyboard pointer

As in the original game, the arrow keys move the pointer on every screen that shows it except the Mission Computer, and `Enter` clicks. The pointer moves in small steps and speeds up while you hold the key. Holding `Enter` while moving drags. This works in every experience mode.

In Enhanced and Modern the keys are remappable (the **Pointer** group on the remap page), and these also work:

Input	Action
Numeric keypad 8, 2, 4, 6	Move the pointer
Numeric keypad 5	Click
Numeric keypad 0	Right click
D-pad / first hat	Move the pointer

The pointer also works at a conversation's Accept/Reject prompt there. In the Navigation Map's Quadrant View the arrow keys walk between quadrants instead; use the numeric keypad or the D-pad to move the pointer there.

7. New Game, Load and Save

The title screen has **NEW**, **LOAD**, **OPTIONS** and **QUIT**.

NEW

NEW starts from the original data in `INITCFG.PAK` or `INITRF.PAK`. If the matching file is missing, NEW is greyed out.

In Enhanced and Modern a start mod can change where and how a new game starts (experimental, see "Start mods" in section 20).

NEW asks for your name and callsign. Press `Esc` on that screen to cancel and go back to the title menu. Righteous Fire's "Load an old Privateer ship?" question has to be answered with `Y` or `N`.

LOAD

Privateer saves use `.SAV`. Righteous Fire saves use `.PRS`.

Put existing DOS saves in the `savegame/` folder of the user-data folder (the launcher's **Savegames** button opens it):

```
savegame/
```

OpenPrivateer reads the original save format.

A damaged save (cut short, or with a broken or missing part, or a file that is not a save at all) is still listed under its name, but choosing it shows "Save game is damaged." and it is not loaded; the log says what is wrong with it. You can delete it from the browser.

SAVE

Use the Mission Computer and choose **SAVE**.

By default, saving under a typed name never overwrites a file. Matching the original game, the name always gets `_NEW` appended (then `_NEW2`, `_NEW3`, ...) rather than reusing the plain name. Typing `PILOT` creates `PILOT_NEW.SAV`, not `PILOT.SAV`; saving under that same name again creates `PILOT_NEW2.SAV`, and so on.

If you would rather keep one file per name, uncheck **Preserve Savegames** in the debug overlay's Experimental section, directly below Vignette (or set `PreserveSavegames=0` in `openprivateer.ini`). Saving then writes exactly the name you typed: typing `PILOT` writes `PILOT.SAV` and replaces an existing `PILOT.SAV`, together with its `.openpriv.json` sidecar. Picking a save in the list and saving again overwrites that save. If a save can not be written, the old file stays as it was. The setting is remembered between runs.

Some OpenPrivateer-only state cannot fit inside the original DOS save format. That state is stored in a matching `.openpriv.json` sidecar while the `.SAV` or `.PRS` remains in the original format.

If you use parked ships, the notification center's messages or other OpenPrivateer-only persistent features, keep the sidecar together with its save.

8. Mission Computer

The Quine 4000 handles saves and pilot information.

Its important pages are:

- **MISSIONS**: active plot mission and accepted side missions
- **FIN**: cash, kills and faction standing
- **MAN**: cargo manifest
- **SAVE / LOAD**: save browser

A pilot can carry up to three side missions. The active plot mission is separate and does not consume one of those slots.

Classic uses the original Mission Computer presentation. Enhanced and Modern can use the alternate OpenPrivateer presentation through the **Quine Model** setting.

Save browser

The SAVE and LOAD browsers list saves alphabetically in every mode.

In Enhanced and Modern the save whose name is in the name field (picked from the list or typed) is highlighted, and these extra controls work:

- mouse wheel over the list: scroll it one save per notch
- `Page Up` / `Page Down`: move one page (10 saves) through the list
- `Home` / `End`: jump to the first or last page
- `Delete`: delete the highlighted save

`Delete` opens a confirmation showing the file, pilot, location, ship, credits and when it was saved (for a damaged save: what is wrong with it). Click the red **DELETE**, or press `Enter` or `Y`, to delete it; click **CANCEL**, or press `Esc` or `N`, to keep it. A controller can click either button with the pointer. Deleting removes the save and its `.openpriv.json` sidecar for good: it does not go to the recycle bin. A read-only save is not deleted; the window says why.

`Delete` also works in the browser opened by the title screen's LOAD, but not when Righteous Fire is importing a Privateer pilot. All five keys can be changed in the controls settings, in the Mission Computer group. In Classic the browser has only the original controls: the `[` and `]` keys (or the arrow keys) and the keypad's arrow buttons.

9. Bases

Bases use the original game's scenes, people and layout.

Move the pointer over doors, people and terminals to see available actions, then click normally.

Depending on the base you may find the Launch Pad, Main Concourse, Bar, Commodity Exchange, Ship Dealer, Repair & Upgrade, Mission Computer, Merchants' Guild or Mercenaries' Guild.

Not every base contains every service.

10. Commodity Exchange

The Commodity Exchange buys and sells cargo. Left-click “Select Items” to buy or sell one unit. Right-click it to buy as many as you can afford and carry, or to sell everything you hold of that commodity, in every experience mode.

Prices and availability depend on the base economy. Each time you land at a base, or load a game, its prices and stock are set anew: what you bought out or sold there is not waiting for you on your next visit, while within one stay the Exchange keeps the prices and the stock your trades left. Cargo consumes hold space. Mission cargo always comes aboard in full when you accept a mission, even when that fills the hold past its capacity; while the hold is over capacity you cannot buy more cargo until you sell or deliver some.

The secret compartment, once obtained through the story, has separate hidden capacity of 20 units, one of them taken by the Ancient Artifact. Tobacco, Ultimate, Brilliance and Slaves you buy, or pick up with the tractor beam, fill the compartment first; whatever does not fit goes into the normal hold. Patrol searches find contraband only in the normal hold. The Exchange sells compartment cargo too: selling one unit takes it from the normal hold first.

11. Ship Dealer and Repair & Upgrade

With normal single-ship play, buying another hull follows the original trade-in model. The current hull and sellable equipment contribute to the trade-in value.

Repair & Upgrade handles ship equipment.

Ships for sale (Enhanced and Modern): click the salesman standing in the middle of the Ship Dealer’s showroom, or press `B` there, to see every ship this dealer sells: the Orion, the Centurion and the Galaxy first, then, with a ship mod that brings ships you fly installed and **Ship Mods** on (in multiplayer the server’s ship mods, whatever **Ship Mods** says), the mod ships this dealer sells. Each row shows the ship’s picture, its name, for a mod ship the stock ship it is registered as, and its price. **BUY** buys it the same way as clicking a ship in the showroom: your ship is traded in (or parked, with multiple ships), the salesman makes his offer and you accept or reject it. The ship you fly is marked YOURS. `B`, `Esc` or CLOSE closes the list. A mod ship may trade in for less than its price, as its mod sets it; the salesman’s offer shows what it is worth.

A mod ship may come with equipment: its row says what (“Comes with …”) and its price includes it. You get it fitted when you buy the ship, onto the empty places of the new ship: a jump drive, scanner or Steltek gun you keep from your old ship stays, and equipment of the mod ship’s that one of them leaves no room for comes off the price instead. Because the price includes the equipment, trading the ship in or selling it parked pays for the hull less what that equipment is worth, and for the equipment on it one item at a time, as on any ship.

In Repair & Upgrade a mod ship shows its own picture where the mod brings one, and you fit guns and launchers on its own positions.

Buy mode lists only what your hull can take, as in the original: each hull has its own shield and engine ceiling and its own turrets, and the Centurion cannot fit the Cargo Expansion. Sell and Repair list whatever the ship carries.

Missiles need a missile launcher and torpedoes a torpedo launcher, and each launcher carries a limited number of rounds (see section 4). Selling a launcher also sells the rounds the launchers you keep cannot carry, each paid at its normal sell price on top of the launcher's: your last missile launcher takes all your missiles, your last torpedo launcher all your torpedoes, and selling one of two launchers sells only the rounds above what the other one holds. When missiles are sold this way, the cheapest go first. A tractor beam does not fire ordnance.

In Classic, entering Repair & Upgrade always starts in **Buy** mode on the first item, matching the original. Enhanced and Modern remember the previous mode and item where possible.

All Equipment (Privateer, Enhanced and Modern, with `RF.TRE` in the game data folder): the dealer also sells Righteous Fire's equipment, with Righteous Fire's prices and pictures: the Fusion Cannon, Gun Cooler, Shield Regenerator, Speed Enhancer, Thrust Enhancer, Isometal Armor, Advanced Droid, and Shield Generator and Engine Upgrade 6 and 7. Shield Generator 6-7 and Engine Upgrade 6-7 fit the Orion only; everything else fits every hull. Without `RF.TRE` the box is greyed out and its hover text says so. In Classic it is greyed out and the dealer is the original's. Weapon mods can change the Fusion Cannon and add guns based on it, as for any other gun.

Equipment you bought this way stays on your ship and keeps working when you turn the setting off or play Classic: Sell and Repair still list it. A Privateer save with this equipment is meant for OpenPrivateer: keep its `.openpriv.json` file next to it. That file tells the Fusion Cannon apart from the Steltek gun, which the save itself stores the same way; without it, and in the original game, a Fusion Cannon loads as a Steltek gun.

A damaged scanner can be repaired in the Software Booth's Repair Mode. The cost depends on the scanner and how badly it is damaged, as in the original game.

12. Multiple Ships and Fleet Selector

Multiple ships are an OpenPrivateer feature.

To use them:

1. Switch to Enhanced or Modern.
2. Enable **Allow Multiple Ships**.
3. Buy another hull.

Instead of trading in the old ship, OpenPrivateer parks it at the current base and makes the new hull active.

On a Launch Pad press:

```
S
```

to open the Fleet Selector.

The active ship is marked as current. Parked ships can be selected or sold. Remote selection of a ship parked at another base is supported.

Multiple ships are not active in Classic.

13. Hired Wingmen

Hired wingmen are an OpenPrivateer feature built on top of multiple-ship ownership.

Enable **Allow Multiple Ships** and **Allow Hired Wingmen** in Enhanced or Modern mode.

A hired pilot flies one of your actual owned parked ships. If the wingman survives, the ship returns with its live equipment state. If it is destroyed, the ship is lost.

Pilots have persistent callsigns and skill. Confirmed kills can improve their skill. Launching with a wingman costs a sortie fee.

A ship assigned to a hired pilot cannot simply be selected or sold as an ordinary parked ship.

Custom callsigns

The callsign pool is stored in:

```
content/wingmen/callsigns.json
```

OpenPrivateer creates the default file if it is missing and does not overwrite an edited version.

The file is a JSON array of non-empty strings. It is read as JSON5, so `//` and `/* */` comments and a comma after the last name are allowed:

```
[
  "Rook",
  "Nova",  // my favourite
]
```

If the file has an error, the built-in names are used and the log says where (line and column). Existing pilots keep their callsigns. Changes affect later hires.

Plain ASCII names are safest with the original game fonts.

14. Flight

Flight follows the original game's own flight model.

Flight includes throttle, afterburner, guns, ordnance, targeting, ITTS, radar, scanners, navigation, autopilot, jumps, docking, tractor beam, turrets, equipment damage and the Repair Droid.

In a turret view (`F5` / `F6`) your flight controls aim the turret instead of steering the ship: the arrow keys, the mouse, a joystick or a gamepad's left stick, whichever you fly with (in Classic, the input device chosen on the Options screen). Up aims the turret down and Down aims it up, as in flight. The ship holds its heading meanwhile.

To jump, fly within range of a jump point and press `J` . Jumping needs a jump drive: without one, `J` does nothing, and a damaged drive can fail to fire, in which case the press does nothing either. As in the original, `J` gives no message when no jump is possible there; only an empty jump-drive tank shows "No fuel" on the Destination page. A full tank holds 6 jumps on every ship, the FUEL gauge shows what is left, and landing at any base fills the tank again for free. All of this works the same way in a multiplayer session.

`[` cycles the left display screen (the only one on ships with one) to its next page and `]` the right one, in the cockpit and in a turret view. Both can be rebound, to a key or a controller button, as **Cycle left MFD** and **Cycle right MFD** in the Controls settings. The page keys themselves (`C` , `G` , `W` , `R` , `M` , `D` , `E` , `V`) cannot be rebound.

The cockpitless HUD

In Enhanced and Modern, `,` (comma) switches between your ship's cockpit and a HUD drawn as thin lines over open space, and back again. It works on every ship, from every view, and can be rebound as **Cockpit / generic HUD** in the Controls settings. Each flight starts in the cockpit, except a mod ship that has no cockpit of its own: that one starts in the cockpitless HUD, and `,` shows its alias's cockpit.

In the front view the HUD shows, across the top, the **AUTO** light (lit when the autopilot can engage), your set speed and actual speed, and the **LOCK** light, which lights red while a missile is homing on you. The crosshair is your cockpit's own, in the same place, so you aim exactly as in the cockpit, and turns red on a lead marker as in every cockpit. At the bottom are two display screens with the radar between a fuel bar (shown with a jump drive) and an energy bar; a bar shows only its filled segments. The left screen shows your shields and armour by default: a bar outside and a block inside for each side (left, right, front at the top, rear at the bottom), each shrinking as it drains, with one tick per shield power level below them. The comm menu, the cargo manifest and the destination page appear there as in the cockpit, and an incoming message shows the speaker's face on a black screen until it ends, when the shield display comes back. The right screen shows the target by default (with a scanner fitted); `G` and `W` bring up the Weapons page there. `[` and `]` cycle the two screens. In a turret view the HUD shows the turret's screen, its energy bar and the crosshair; the side and rear views show only space. Switching resets the screens' pages, except that a message being received stays up. The message line keeps its place (see "Message line" for moving it).

Two settings in the debug overlay's GAME section change the cockpitless HUD's look, in Enhanced and Modern: **HUD Opacity** (Auto, or 10% to 100%) fades its lines, bars, lights, radar circles and screen corners, while the display screens' pages, the speed readout, your cockpit's own crosshair and the comm face stay solid; **HUD Brackets** (Auto, On or Off) shows or hides the corner marks around the two display screens. Auto leaves both to the HUD style (full strength and corners on unless a style says otherwise). Both apply at once and are kept between sessions.

Your own HUD style. A file `content/hud/hud_style.json` in the user-data folder (beside `content/mods` , see section 2) restyles the cockpitless HUD on every ship: move or recolour the bars, the radar, the lights and the speed readout, hide parts, draw a simple line crosshair instead of the cockpit's,

set the opacity, and choose which pages each display screen shows, for example the Weapons page on the left screen:

```
{
  parts: {
    left_mfd: { pages: ['weapons', 'shields'] },
    right_mfd: { pages: ['target', 'damage'] },
  },
}
```

Each screen then cycles through its own list with `[` or `]` and starts a flight on the first page of its list; a page's key (`G` or `W` for Weapons) brings it up on the screen that lists it. Listing `comm` on the right screen moves the comm menu and incoming messages there. A mod ship may bring a style of its own; your file wins over it wherever both set something, and the two settings above win over both. A file with a mistake is left out and the reason is written to the log. Every member and its limits are listed in `doc/formats/ship_mod_format.md`, "The generic HUD's style". The file is read when the game starts and again with **Reload ship mods** in the debug overlay's MODS section (with **Auto reload** on, every time you save it; in flight the game must be paused).

Shields

`S` steps the shield generator's power level down and `Shift` + `S` steps it back up, one level at a time, in every experience mode, as in the original. A lower level holds less charge per facing but draws less energy; level 0 switches the shields off and empties them. Pressing `S` again at level 0 switches them back on at full power, and they recharge from empty. `Shift` + `S` stops at full power. A smaller generator has fewer levels; one with only on and off switches between the two on every `S`.

Scanner

Targeting requires a working scanner.

With no scanner fitted, the radar is absent and scanner-dependent targeting functions are unavailable. There is no hidden free scanner.

A selected target reads "Unknown" on the Target display until the scanner identifies it; a better scanner identifies sooner. Steltek ships and Righteous Fire's Salthi always read "Unknown".

The Steltek drone

Only the boosted Steltek gun damages the Steltek drone. Every other gun, the Steltek gun before it is boosted, and every missile and torpedo hit it without effect.

Tractor beam

Select a cargo canister or an ejected pilot with `T` first; it does not need to be locked. With nothing selected, or with a ship, base or planet selected, `Enter` does nothing. A tractor beam in a turret is switched on and off with `Enter` from that turret's view (`F5` / `F6`). A tractor beam on one of the ship's own launcher hardpoints is selected with `W` from a forward view - it comes after the missiles and torpedoes and the Weapons display reads `W: Tractor Beam` - and then switched on and off with `Enter` . A beam keeps pulling the object it was switched on against, even if you change views or select something else. It switches itself off without a message when that object is gone, more than 10000 units away, does not fit in the hold, or the ship runs out of energy. A beam on a launcher hardpoint is drawn from that hardpoint, below the nose.

As in the original, several tractor beams can run at once and each one is drawn. Each beam that pulls an object in adds it to the cargo hold, so two beams on one object give you two of it.

Autopilot

Autopilot follows the original instant-travel behavior. It drops you out at the edge of the destination's own area, so the ships waiting there are already present when you arrive: about 12,000 m from a jump point, further out from a base or planet (17,000 m or more), and nearer to some smaller nav points. It drops out early when the route enters another encounter zone on the way, including asteroid fields. Autopilot reports "Navpoint near" once you are already inside the destination's area.

Message line

Comm messages, mission progress and other short notices appear on the cockpit message line, a bar at the bottom of the screen. **Plaque Position** in the GAME section of the settings overlay moves it, in every cockpit and every experience mode, Classic included:

- **Bottom** (the default): at the bottom of the screen, as in the original.
- **Top**: at the top of the screen. While the missile camera's "Missile Camera On/Off" message is up (`F7`), the line sits just below it.
- **Off**: the message line and the missile camera message are not shown. Spoken comm messages are still heard; a message that is only text is not seen.

The setting also places a base's greeting caption shown while you land. The "Game Paused" message always shows in the middle of the screen. The setting is remembered between runs.

Pause

Press `P` in flight.

15. Navigation Map

The Navigation Map uses the original sector and quadrant data.

It shows systems, nav points, bases, jump points and mission objectives. As in the original, only jump points are blue circles; bases are green squares, and pirate bases and empty nav points are green circles. Story-locked systems and routes remain hidden until the game makes them available. Once a

story jump point is shown (the Kilrathi systems beyond Rygannon), it can be flown through like any other jump point.

Active mission targets are highlighted.

The optional **Open Universe** setting exposes additional universe content and is not part of Classic behavior.

16. Guilds and Random Missions

The Merchants' Guild, Mercenaries' Guild and Mission Computer provide the original non-plot mission types:

- Cargo
- Patrol
- Scout
- Defend
- Attack
- Bounty

Guild fees are:

Guild	Fee
Merchants' Guild	1,000 credits
Mercenaries' Guild	5,000 credits

Both guild computers show the original **Scanning … records … Please stand by…** stage while generating a fresh batch of offers.

Notification center

In Enhanced and Modern, every mission the game pays or fails leaves a message in your notification center. On a Launch Pad press:

```
M
```

to open it. After a landing that completed or failed a mission it opens by itself once you are on the Launch Pad.

- A **completed** mission shows what it paid. A **failed** one shows the reason the game decided, for example **Nav point 2 not reached**, **Target not destroyed**, **Not delivered to Liverpool** or **Cargo missing (10 x Grain needed)**, and whether the mission cargo was taken back. Each message also shows where it happened, the mission number and the time.
- A mission you remove with the Mission Computer's **REJCT** button leaves no message.
- A mission whose destination base was destroyed fails on your next landing, with the reason **<base> was destroyed**, and its mission cargo is taken back. Mission boards never offer a destroyed base as a destination.

- Loading a save whose ship or equipment came from a mod that is no longer installed leaves a message naming what your ship carries in its place.
- An unread message has a bold title. **MARK READ** marks it read.
- The red **DELETE** button removes a message after you confirm.
- **MARK ALL READ** and the red **DELETE ALL** (after you confirm) act on every message.
- Newest messages are at the top; scroll the list with the mouse wheel. **M**, **Escape** or **CLOSE** closes it; Escape first cancels an open confirmation.
- The key can be changed under **Notification center** in Remap Controls. It has no controller button by default; you can assign one there.

Messages are saved with your game. Classic keeps recording them without showing them, so they appear once you switch to Enhanced or Modern. The notification center is not available in multiplayer.

Mission progress in flight

In Enhanced and Modern, the cockpit message line (see “Message line” in section 14) tells you when you complete a mission objective in flight. It shows the objective as the Navigation Map lists it, for example:

- **GO TO DELTA, NAV 2: COMPLETE** when you reach a patrol nav point
- **DEFEND PALAN, NAV 1: COMPLETE** when a nav point is clear of hostiles
- **DESTROY RHOMBUS: COMPLETE** when the target is destroyed

Once everything that needs you in flight is done, it shows **OBJECTIVES MET - LAND AT LIVERPOOL** (or just **OBJECTIVES MET** when the mission names no base). You are paid when you land. Travel legs and cargo deliveries show nothing in flight. With more than one mission carried, each message starts with the mission’s number as the Navigation Map pages show it, for example **MISSION 2 - …**. Turning on Invulnerability or Unlimited Ammo in Options voids every mission you carry, and the message line says so.

Each message stays for 3 seconds, one at a time. A comm message always comes first; a mission message it interrupts is shown again afterwards. They are not shown in a turret view. The Navigation Map’s objective lines also change colour as you complete them.

Untick **Show Mission Feedback** in the GAME section of the settings overlay to turn them off. They are not shown in Classic or in multiplayer.

17. Music and Sound

OpenPrivateer supports several music backends, including OPL2, MIDI with an external SoundFont, OS MIDI and CD-style replacement music. Availability depends on the platform.

Volume

The settings overlay’s VOLUME section has three sliders, each from 0 to 127 and remembered between runs:

- **Music**: the music, with every backend.

- **SFX**: sound effects: engines, guns, missiles, explosions, cockpit and cutscene sounds.
- **Speech**: every spoken line: base conversations, in-flight comm messages and cutscene voices.

The three are independent of one another, and a change is heard at once, also on music or a line that is already playing. The Options screen's **Music** and **Sound F/X** switches still turn music, or sound effects and speech, off altogether.

SoundFont

Select a `.sf2` file in the launcher's **Music** section or from the SoundFont setting when using the corresponding MIDI backend. The path is stored in `openprivateer.ini`. Without one, OpenPrivateer looks for `openprivateer.sf2` in the user-data folder, then next to the program.

Replacement music

The `music/` folder accepts your own music files in place of the game's built-in music. WAV, MP3 and FLAC are supported. To hear them, choose **CD (your own files)** in the launcher's **Music** section, or **CD** as the **Music Backend** in the settings overlay.

Each piece of the game's music has a slot: a subfolder and a two-digit number. Name your file with that number, an underscore and anything you like, for example `basetune/04_mining.mp3`. The subfolders are `basetune` (the music at each kind of base), `basetune2` (Righteous Fire's Gaea), `combat` (the music in space, by situation), `combat/000` (the short bridges between two pieces in space), and `opening`, `opening2` and `victory` (the intro, Righteous Fire's intro and the victory movie). A base's music, a movie's music and the music of a lasting situation in space repeat the whole file; short pieces play once. A slot without a file stays silent while this backend is chosen.

The generated `music/README.txt` lists every slot number. It is rewritten each time the game starts, so it always matches the installed build.

Phase torpedo decode music

Put one WAV, MP3 or FLAC file in `music/phase_decode/` to hear it while a phase torpedo decodes, for example `01_decode.wav`. It works with every music backend. When a decode starts, the other music stops and your file plays once from its beginning; it does not repeat, and the decode completing does not stop it. It stops, and the other music comes back, when you fire the torpedo or the decode is lost (turning away, the target leaving the screen, another target or weapon, leaving the cockpit view, a jump, landing). The next decode plays it from the beginning again. With several files in the folder, the first in name order plays. It follows the Music volume and stays silent with Music off. The folder takes the music only; a sound when the decode completes comes from the weapon mod that makes the phase torpedo, if it brings one.

Music of an interior mod

A base with an interior of its own (section 20) can bring its own music as WAV, FLAC or MP3 files: one piece for the whole base, and pieces for single rooms, such as a bar with music of its own. It plays with every music backend, follows the Music volume and stays silent with Music off; Sound F/X does not affect it. Going between two rooms with different pieces fades one into the other; rooms sharing a piece let it play on. The shops and terminals you open from a room keep the room's music. Such a base

can also play music of its own over the landing and takeoff movies; on landing it can run straight on into the base's music.

18. Graphics

OpenPrivateer renders the original game at its native logical resolution and scales it for modern displays.

Available settings include CRT filtering, image adjustments, scaling, texture filtering, pixel scaling and splash font selection.

Classic keeps the original's look, including its quirks. Enhanced and Modern apply some presentation improvements, such as keeping round sprites round while the ship rolls.

Hull Atlas

Hull Atlas in the debug overlay's MODS section (at its end, or `HullAtlas=Auto`, `On` or `Off` in `openprivateer.ini`) decides how the pictures of ships from ship mods reach the graphics card. With Auto (the default) or On, each ship's pictures travel together as one large image, so a mod ship appears for the first time without a stutter. Off sends every picture on its own, which takes a little less graphics memory (the combined image holds about 6 to 20 percent more) but costs a short pause each time a new mod ship appears. Turn it off only if a system short of graphics memory has trouble with many mod ships at once. The ships look the same either way, and the original game's ships are not affected. A change applies to ships that appear afterwards; ships already shown keep theirs until the next star system. An unknown value in the ini file reads as Auto.

19. Wallpaper

The `wallpaper/` folder accepts custom artwork for the Privateer / Righteous Fire selector.

Use filenames beginning with:

```
01_
```

for Privateer and:

```
02_
```

for Righteous Fire.

PNG, JPG and BMP are supported. See `wallpaper/README.txt` for the current details.

20. Content Overrides and Translations

OpenPrivateer can load replacement files from the filesystem without changing the original TRE archives.

The basic layout is:

```
content/

    priv/

    rf/

    i18n/

    mods/universe/

    mods/weapons/
```

`content/priv/` and `content/rf/` mirror paths inside the corresponding original archive.

Each universe mod is a folder in `content/mods/universe/`, named after the mod, holding any of the files below (for example `content/mods/universe/haven/systems.json`). The folder name becomes the prefix of the mod's keys (`haven/new-haven`) unless a file sets its own `"mod"`. The name `openprivateer` belongs to the game's own content and cannot be used for a mod. File names must be written in lower case exactly as shown (`systems.json`, not `Systems.json`); a file with another spelling is not read and the log says so. Files placed directly in `content/mods/universe/` also work, as one more mod without a folder. Several mods can be installed at once: the files directly in the folder load first, then the mod folders in name order, and each kind of file is read for all mods before the next kind, so one mod can use another mod's factions, quadrants and systems. When two mods want the same thing (the same key, the same empty place on the Sector Overview), the one loaded first gets it and the other mod's file is not loaded. Each mod may have its own `filter.json`; what they hide adds up. The Open Universe setting turns all universe mods on or off together.

`content/mods/universe/systems.json` adds systems and jump lines to the Open Universe. The file is JSON5, so comments, trailing commas, unquoted field names, single-quoted strings and hex numbers are allowed, but each field may appear only once per object. Invalid files are rejected as a whole and the log names the file and the field, or the line and column of a typo; the other files still load. Delete a file to turn it off.

`content/mods/universe/sectors.json` adds new sectors to the Open Universe map, each in one of the five empty places of the Sector Overview, with up to four new quadrants that `systems.json` can fill.

`content/mods/universe/filter.json` hides sectors and systems from the Open Universe map. Gemini is always shown. Hidden systems still exist: the filter only changes what the map shows.

`content/mods/universe/factions.json` adds factions with their own map color and name (shown as "Faction" in a selected system's info panel). A system in `systems.json` can name one in its optional `faction` field.

`content/mods/universe/navpoints.json` adds bases, planets, stations, plain nav points (empty space, an asteroid field) and jump points. Added bases can be flown to and landed at. A mod base can reuse the interior of a similar original base while keeping its own location identity. A nav point can be made a no-weapons-fire zone. A mod base can also give its own name for the Communications MFD contact list and its comm messages, and its own type line on the Mission Computer under its name (for

example “Carrier” and “Confed Carrier”); without them it shows those of the original base it borrows from, except that in Enhanced and Modern its contact row shows its own name. That type line is also the base’s type in the navigation map’s “Bases :” list instead of its category; a type too wide for the list ends in “…”.

A base or station can also be a **carrier**: a station from a ship mod stands on the nav point, and the base is aboard it. You see, target and hail it as a ship, and you dock and land on it like at any base: inside, you save, repair, trade and launch with the services the base offers, and you launch next to the carrier. A carrier can have its landing deck off its centre, at the stern for example: the landing zone, the base’s contact on the Communications MFD and your launch point are then at that deck rather than at the middle of the ship. As at any base, you land by pressing `Esc` inside its landing zone or sending the deck’s docking request (with **Hail To Land** on, within 2000 units of the deck; section 4). A carrier stays on its spot when you ram it, though the ram still damages both ships. Docking is open only while the carrier is not hostile to you; fly into its landing zone while it is hostile and the message line shows “Docking refused”. If the carrier is destroyed, its base stays closed until you next enter the system, when the carrier is back. A mod can make a carrier’s base **destructible**: then destroying the carrier (or, depending on the mod, only your own kill of it) destroys the base for good. It is gone from space, perhaps leaving a wreck you cannot target or land at, and the navigation map shows its nav point as “<name> (destroyed)”. It stays gone in your saved game; some mods bring it back after a number of landings elsewhere, others never. Destroying such a base moves your standing as the mod sets for that base (by default the faction that flies the carrier trusts you less), and it counts as one kill. Only a weapon a mod makes a base killer destroys such a carrier with one shot; a one-shot ship killer only damages it like any gun. A mod can also make a planet destructible: only a planet-killing weapon destroys it, and then the planet and its base are gone from space in the same way. A carrier needs **Ship Mods** on (Enhanced or Modern, single player); without it the carrier’s base cannot be docked at, and a game saved aboard it loads at the original base it borrows its interior from.

Some carriers and added bases **answer when you ask to land** (a mod gives them a comm profile). Inside the landing zone their docking list on the Communications MFD starts with “Requesting immediate automatic docking”, then a request line (“Request landing clearance.” or the mod’s own words). Send the request line and the base answers: cleared, or denied, for example while enemy ships are near it. The answer shows on the message line, or, when the mod gives the base a voice and sound effects are on, you hear it spoken instead. Such a base may also show a face of its own on the comm display, which talks while it speaks. Once cleared you land by flying in closer than 1000 units to the deck or by pressing `Esc`; leaving the landing zone cancels the clearance, and a denied request can be sent again at once. With **Hail To Land** on, `Esc` at such a base sends the request first if you are not cleared yet. “Requesting immediate automatic docking” (key 1 on the list) lands at once, whatever the base answers; at a carrier with **Hail To Land** on it is listed only within 2000 units of the deck, and the request line is key 1 until then. Such a base may also list lines of its own on the Communications MFD; it never answers them, and the page closes when you pick one. A carrier that has turned hostile may tell you so itself when you fly into its landing zone. Such a base may also **greet you as you land** there (“Welcome aboard, sir.” or the mod’s own words), however you land: during the landing movie and on the landing pad you hear it spoken, when the mod gives it a voice and sound effects are on, or read it for a few seconds on the bar at the foot of the screen. It greets you once per landing, never when it refuses you.

An added base or station can also have an **interior of its own** from an interior mod (`content/mods/interiors/<mod>/interiors.json`, or a pack’s `interiors/` folder): new pictures for its landing pad, concourse, bar or showroom, new rooms, and doors that lead elsewhere. An interior either

builds on an original base (every screen the mod does not change is that base's) or is entirely its own, a new kind of base whose doors open the shops of the original base it stands for. Its rooms can move (flickering lights, smoke, figures that idle and turn to look when you point at them, a train that rides off when clicked), play a sound of their own while you are in them, have a barman who tells rumours when you click him, and sell ships from their own "Purchase" spots, as a Ship Dealer's showroom does. People in its rooms can talk to you when you click them, as the original's bar characters do: their lines show as text (they have no voice), a click or Space moves on to the next line, Esc jumps to the question, and Accept or Reject answers it. What you answer can matter later: a character may remember a job you took, pay or charge you, or be gone the next time you come, and doors can appear or close. This is kept with your saved game. The Commodity Exchange, the guild offices, the Mission Computer, Repair & Upgrade, the Software Booth and the Research Computers are always the original screens. Their Exit takes you back to the room you came from, and so does Escape in Enhanced and Modern (on a landing pad, Escape takes you into the base). A door to a service the base does not offer is not there. A room's sound is a sound effect: it plays with Sound F/X on. A room may also silence the base music while you are in it, so that only its own sound plays; the music comes back when you go to a room that has it. Such a base's sound can carry on into the original screens you open from its rooms (the bar, the Mission Computer and the others), or change to one of their own while they show. Such a base can bring its own music (section 17), for the whole base or for single rooms. Such a base can also change the landing and takeoff movies: you may see your ship approach a picture of the base's own in place of the usual one, hear music of its own over the landing or the launch, or land or launch without a movie at all. Interiors show in Enhanced and Modern with the Open Universe on and **Interior Mods** on (the debug overlay's MODS section, or `InteriorMods=0` in `openprivateer.ini`; on by default), never in Classic or in multiplayer; with them off, or without the mod, the base shows the original interior it borrows. The change applies at your next landing or load. A game saved at such a base comes back there with the interior, and with what its characters remember about you (also when the mod is missing for a while: once it is back, so are they). Starting the game with `openprivateer --interiors-report` writes the loaded interiors to the log and the console; the generated `content/mods/interiors/README.txt` has an example, and the asset export (section 25) writes every original base's interior in this format to start from.

`content/mods/universe/zones.json` adds asteroid fields to systems. A zone can use a position or a nav point to sit on, a size and a density (`"sparse"`, `"medium"`, `"dense"` or 1 to 7 rocks).

`content/mods/universe/encounters.json` adds random encounters: for a nav point or zone, a list of possible groups with their chance, number of ships, faction, ship type, skill and morale. Each time you enter the system one group per table is picked, and its ships appear when you come near that point, as in the original systems. Several tables, also from different mods, can share a nav point or zone: each picks its own group, all of their ships appear there together, and the log notes each extra table. A table can also name one of the original Gemini systems and one of its nav points as the Navigation Map shows it (`"nav": "Nav 1"`): its ships appear there beside the original's, in Enhanced or Modern with Open Universe on and outside multiplayer, and in a multiplayer session whose server runs the mod; Classic keeps the original's encounters alone. Such a table can also take away or change the original's own groups at that nav point (`replace`, `remove`, `modify`): a group that was taken out keeps its chance, so when it comes up that nav point has fewer ships or none. Groups that run a script, such as the militia's contraband searches and their escorts, always stay. `openprivateer --encounters-report` writes every original group of every Gemini system with its record number to the log and the console, and what the loaded mods do to it; `--encounters-report=Humboldt/Troy` lists one system. The game starts normally afterwards.

The generated `content/mods/universe/README.txt` contains the current formats and built-in examples.

Mod packs, load order and disabling mods

Every kind of mod (universe, weapons, items, start, mission, ship and interior mods) can be installed in two ways, which work the same:

- one folder per kind, as described in each section here, for example `content/mods/weapons/rebalance/weapons.json`;
- a **mod pack**: one folder holding all of a mod's parts, in `content/mods/packs/`, for example `content/mods/packs/rebalance/weapons/weapons.json` and `content/mods/packs/rebalance/items/shipstuff.json`. A pack is the easiest way to share a mod: zip its folder.

The folder name is the mod's name. If a pack and a per-kind folder have the same name, the pack is used. A mod may come with a `mod.json` that gives its full name, version, author and description, and which other mods it needs or does not work with. A mod that needs a mod you do not have, or that does not work with a mod loaded before it, is not loaded, and the log says why.

Mods load in alphabetical order of their names (a mod after the mods it needs). To choose the order yourself, or to disable a mod without deleting it, use the launcher's **Mods** section (below), or write `mods.json` in the settings folder (next to `openprivateer.ini`) by hand:

```
{ "version": 1, "order": ["base-pack", "rebalance"], "disabled": ["test-mod"] }
```

The mods listed under `order` load in that order; mods not listed load after them. The mods listed under `disabled` stay disabled. Changes apply the next time you start the game. Start the game with `--mods-report` to see which mods were found, their order, which are disabled or not loaded and why; the same is written to `logs/mods_status.json` at every start. The generated `content/mods/README.txt` sums this up.

Managing mods in the launcher

The launcher's **Mods** section lists every installed mod in load order. Its title says how many are enabled. **Mods are disabled in Classic mode**: they apply only in Enhanced and Modern. A line under it lists the **Enabled features** (Weapons, Items, Missions, Ships, Interiors, New Game, Open Universe), set with the debug overlay's switches.

- The first row, **Built-in Universe**, is OpenPrivateer's own Open Universe content. Its box is the same setting as the debug overlay's **Built-in Content**.
- **Loose universe files** (the files directly in `content/mods/universe/`) are always enabled and always load first.
- Every other mod has a box to enable or disable it, and **Up**, **Down**, **Top** and **Bottom** to change its place in the load order. A mod later in the order wins where two mods change the same thing.
- **Version** is the mod's version from its `mod.json`; **Hash** is the first 8 characters of its content hash, which changes whenever one of the mod's files changes, even when the version does not (hover over it for the whole hash).

- Click a mod's name for its details: author, description, web address, where its files are, its content hash, files in its folder that are not part of the mod, and the warnings and errors of the last start.
- **Status** shows what the last game launch did with the mod: **loaded**, **partly loaded** (one of its files was rejected), **not loaded**, **new since the last start**, or **Not checked yet** before the first launch. Until then the section says **Mod status will appear after the next game launch.**
- A mod that needs another mod, or does not work with one, shows the problem under its row with a button that fixes it, for example **Enable base-pack** or **Move after base-pack**. A mod whose `mod.json` lists files that differ from the files on disk says which ones.
- **New Game Mod** picks the start mod used when starting a new game; **(automatic)** uses the only one when there is exactly one.
- **Open Mods Folder** opens `content/mods`; **Refresh** reads the folders again after you added or removed a mod.

Every change is saved at once (the order and the disabled mods in `mods.json`, the start mod and Built-in Universe in `openprivateer.ini`) and takes effect on the next game launch. Some mod settings can also be changed in-game, in the debug overlay: Weapon Mods and Item Mods at once, Mission Mods at the next mission board, Ship Mods the next time you enter a star system, Interior Mods at your next landing or load, Init Mods and Init Mod at the next New Game. A mod list that cannot be read is shown with its error and a button that sets it aside as `mods.json.bak`; until then nothing in the list can be changed.

In a running game, the **This run** list in the debug overlay's MODS section (click **This run** to open it) shows what this start did with each mod and how many errors and warnings it had, and **Mod manager (restarts)** restarts OpenPrivateer into the launcher with the Mods section open. Progress since the last save is lost.

The overlay's right-hand pane is the **Mods log**: every message the mods gave since the game started, warnings in orange and errors in red, long lines wrapped so they can be read in full. Above it are the number of warnings and errors, **Problems only** to show just those, and **All log** to show the game's whole log instead.

Reloading mods while you play. Below that list, **Reload ship mods** reads your ship mods again without restarting, and **Reload interior mods** does the same for base interiors. This is meant for mod authors trying out changes:

- Docked, at a base screen or in a menu, **Reload ship mods** applies every change to your ship mods. Ships you meet in space change from the next time you enter a star system. If the ship you own loses its mod hull in the reload, it becomes the stock ship that hull was based on, as when a save is loaded without the mod.
- In flight, **Reload ship mods** works only while the game is paused (P). It applies the cockpit, launch pad and Repair & Upgrade pictures and the Weapons MFD at once. Flight handling and hardpoints change at your next launch. A change that adds, removes or renames a ship, or changes a ship's mounts, hold, cockpit file or dealer details, is refused in flight: land and reload again.
- **Reload interior mods** works when you are not in flight. Docked at a mod interior, the visit starts again on the scene you were on, with the new pictures, hotspots, music and sounds.
- A reload waits while a movie, a conversation or a click script is running, and runs as soon as it ends. In flight it waits until you pause.

- A reload is refused in a multiplayer session or while connected to a server. It is also refused when the reload would break something only a restart can set up again. A ship that an encounter table or a carrier base uses is one example; an interior that a nav point names is another. The overlay line under the buttons and the Mods log say why.
- If a mod no longer loads (for example after a typing mistake in its file), nothing changes and the error is shown.
- **Auto reload** (off every time the game starts) reloads by itself about a second after you save a change to a ship or interior mod file.
- Classic plays without mods, so the buttons are disabled there.

Reporting a bug with mods installed: name each mod's version and hash. The log in `logs/` lists every mod at every start, one line each, for example `mod 2. tayla, version 0.C 2026-10-21, hash 1a2b3c4d, loaded, layers ships,universe`; the launcher's Mods section shows the same version and hash. Two copies of a mod with the same hash have the same files. When a multiplayer join is refused because the mods differ, the log names your copy's version and hash and the server's version and hash for each mod that differs; the launcher's **Check Server** shows the same before you join.

Weapon mods (experimental)

For every kind of mod (weapons, items, start, mission and ship mods below, and the universe and interior mods above), a mod folder may not be named `openprivateer` or `packs`: those names belong to the game, and such a folder is skipped. The mod's file must be named in lower case exactly as shown (`weapons.json`, not `Weapons.json`); a file with another spelling is not read and the log says so. Universe mods are the exception: their files are found in any letter case. Pictures and sounds a mod names inside its file are found by the name written there. A mod's file may be up to 4 MiB; each picture or sound up to 8 MiB, and each picture up to 2048 by 2048 pixels. A larger file is not used, and the log names it and the limit.

A weapon mod changes the guns. OpenPrivateer comes with none: you add them yourself, and without one the guns are the original's. Each mod is its own folder with a `weapons.json` in it, for example `content/mods/weapons/rebalance/weapons.json`; mods load in alphabetical order of their folder names, and a later mod's value for the same gun and stat wins. A mod can change a gun's damage, time between shots, energy per shot, bolt speed and lifetime (and with them its range), its wear limit and its three-letter name on the Weapons display; a changed original gun changes for every ship that carries it, computer-flown ships included. Named stat sets let several guns share one baseline and adjust it. A weapon mod can also set a gun's name, prices, the ships the Ship Dealer sells it for and the bases that sell it; where a Ship Dealer mod (below) sets the same thing for that gun, the weapon mod's value is used and the log says so. The `README.txt` in `content/mods/weapons/` shows a first mod, and the modding documentation that comes with the source describes every field with worked examples.

A mod can also add new guns. Each new gun is based on an existing one, and the Ship Dealer sells it under its own name (in the selected language when the mod gives one) and price, after the original items, on the ships the mod lists; buying, selling, repairing and fitting it on a turret work as for any gun. A new gun based on the Steltek gun is kept like the Steltek gun unless the mod says otherwise: it takes no wear and cannot be sold. Ships you meet in space do not carry new guns; your hired wingmen fire them on your own ships. A mod can give a gun its own pictures (the bolt, the Weapons display icon, the dealer's icon, its picture on each ship's hardpoints, turret bays included), its own lead-pip colour and its own firing sound. Pictures are PNG files, drawn in the game's own colours; sounds are WAV or VOC files.

The Laser's bolt and a beam weapon's beam (below) are drawn as lines, and a mod can give each gun's line its own colour and thickness, and the Laser's bolt its own length; only the look changes, not where the shot hits. Every ship firing that gun shows it, other pilots' ships in a multiplayer session included.

A mod gun can be a charge weapon: hold the fire button to charge it and release to fire one shot, stronger the longer you charged. Charging uses energy while you hold; releasing too early, switching views or running out of energy loses the charge. Some charge weapons fire on a quick press as well: let go before the minimum charge and the weapon finishes charging to its minimum by itself and fires a minimum shot; keep holding and it charges on as usual. A small bar above the Weapons display's `G:` line shows the charge. A weapon with a charged sound plays it the moment the charge is full, whether it then waits for you to let go or fires by itself. Ships flown by the computer fire charge weapons too: they charge while they have a target in range and fire when the charge is full and the gun points at it. A turret of your own ship that you are not sitting in does not fire a charge weapon. A charge can take up to 20 seconds, and longer when a mod allows it.

A mod weapon can be **lethal**: one hit that reaches a ship destroys it, whatever its shields and armour, and that includes your ship. A phase shield (see "Ship mods" below) stops a lethal hit unless the weapon can go through phase shields. An enemy capital ship can carry such a gun, a cannon that charges for a long time and then fires one huge shot that destroys any ship it hits. When a computer-flown ship within about 5000 units of you is charging a weapon, you hear the weapon's charging sound, if the mod gives it one; the shot sounds like any other ship's shot. Stay out of its gun's arc, or out of its range, while it charges.

A weapon mod can make your torpedoes **phase torpedoes**, the one weapon of a fighter that gets through a capital ship's phase shield. To use one:

1. Select the torpedo with `W`, target the phase-shielded ship with `T` and lock it with `L`.
2. Keep the ship in front of your nose, inside what the front view shows, within the range the mod sets (12000 unless it says otherwise), and keep the nose steady: small corrections are fine and rolling is fine, but a hard turn of more than half a second interrupts the decode. While the torpedo decodes, a red reticle (an oval ring with a cross through it) circles the target, starting well away from it and closing in as the decode builds (20 seconds unless the mod says otherwise); rolling or pitching your ship turns it around the target. An interrupt makes the reticle disappear, and the decode starts again (or, with some mods, waits where it was).
3. When the reticle reaches the middle of the target, the reticle and the target box flash and the missile-lock diamond appears (with a sound, if the mod brings one). From then on you can manoeuvre and close in, as long as the ship stays in front of your nose: turning away from it loses the decode (the reticle disappears), and so do changing or unlocking the target, selecting another weapon with `W`, taking a turret's seat (`F5` / `F6`), jumping and landing. Looking left, right or back, or through an outside camera, does not matter: only where your ship points counts.
4. Press `Enter` at the range you like (about 3000 is typical): the torpedo goes through that ship's phase shield, and with most mods it homes on the ship. Each torpedo needs its own decode.

The typical attack: start the decode at long range, outside the capital ship's gun range, then fly in toward it without turning away and fire at close range.

A torpedo will not launch at a phase-shielded ship before it has decoded ("Torpedo not decoded" on the message line). At a fighter, or any ship without a raised phase shield, the torpedo launches and flies as an ordinary torpedo. A mod can give a decoded torpedo its own speed and flight time, often slower

than an ordinary torpedo and much longer, and its own damage, so a decoded torpedo can hit far harder than an ordinary one while the torpedoes of other ships stay ordinary. Phase torpedoes work in single-player only; computer-flown ships launch ordinary torpedoes.

A weapon mod can also make a missile's lock **take a moment**, as in Wing Commander 2 (Heat Seeker, Image Recognition and Friend or Foe missiles; a Dumb Fire never locks). With such a missile selected and its usual lock conditions met (the Heat Seeker behind the target, the target locked with `L` for the other two), keep the target in front of your nose: the same red reticle circles it and closes in, quicker than a torpedo's decode (under a second to a few seconds, as the mod sets). When it reaches the middle of the target, the reticle and the target box flash and the missile-lock diamond appears: the missile is locked. Losing the conditions or turning away from the target breaks the lock (a mod may allow a short slip, or keep the progress), and so do changing the target or the missile and taking a turret's seat; looking around, as for the torpedo, does not. Every missile you launch uses the lock up, so the next one locks again from the start. A missile fired before it is locked flies straight without homing; with some mods it does not launch at all and the message line says "Need lock". Timed missile locks work in single-player only; computer-flown ships lock the way the original game's do.

A weapon mod can also give a missile or the torpedo **lock sounds**: a sound that repeats while a lock builds, one when the missile locks (for a missile that locks at once, when the lock diamond appears) and one when the lock is lost. Launching a missile uses its lock up without the "lost" sound. The original game has no lock sounds, so without such a mod a missile locks silently. Lock sounds also play in multiplayer.

A weapon mod can also change the missiles, the torpedo and the two launchers: their damage, how long a round flies, the range at which the missile-lock diamond appears, how quickly a launcher reloads, and how a round flies: its top speed, how fast it gets there, and how sharply a homing missile can turn. It can also rename a missile or the torpedo on the Weapons display (up to 8 characters); the name the Ship Dealer lists comes from an items mod instead. The change applies to every ship that carries them. New missile types cannot be added.

A weapon mod can also give a missile or the torpedo a **limited burn time**: once its fuel is out it can no longer steer or speed up and flies straight on at the speed it had until it expires, so a missile fired from too far away may coast past its target. The missile-lock warning goes out once an incoming missile's fuel is spent. A mod can also make a missile leave the launcher already moving, at a set speed along its nose, with your ship's forward speed or your ship's whole velocity added. Computer-flown ships fire such missiles only at targets they can reach under power. These missiles work the same in multiplayer.

A weapon mod can also change the course a homing missile takes: it can aim ahead of a moving target, at where the target will be (**lead**), so it meets a crossing ship from the side and sooner, even when it cannot turn sharply; or come in behind it (**lag**), swinging onto the target's tail and closing from behind, which takes longer and can miss a fast ship crossing in front of it. A missile with neither steers straight at where its target is. Computer-flown ships fire these missiles too, and they work the same in multiplayer.

A weapon mod can also give a missile a **narrow seeker**: it sees its target only within a cone in front of it. A missile fired at a target outside that cone flies straight ahead unguided, and one whose target slips out of the cone in flight, for example when the target turns hard across its nose, stops homing and flies straight on; the missile-lock warning goes out when that happens. Some such missiles pick the same target up again when it comes back in front of them; they never switch to another ship. Computer-

flown ships fire these missiles too, and they work the same in multiplayer, where every pilot sees a missile stop and start homing at the same moment.

A weapon mod can also make a missile **guided by your ship**: it homes only while you keep its target selected and locked with `L` and in front of you, within a cone around your nose. Turn away or release the lock and the missile flies straight on; bring the target back in front of you with the lock on and it homes again. Select another target, or lose your ship, and the missile flies on unguided for good. The missile-lock warning shows only while such a missile is homing. Computer-flown ships fire these missiles too and guide them at the ship they are fighting, but they keep maneuvering while the missile flies, so theirs often go astray. They work the same in multiplayer: the server guides the missile from the target you have selected and locked.

A weapon mod can also give a missile or the torpedo a **warhead**: it explodes when it passes close to its target, or where it hits, and the blast hurts every ship near it, less the farther away a ship is. Your wingmen and other bystanders close to the blast are hurt too, and so is any ship nearby that you would rather not anger: a blast counts as your shot. Your own ship is spared by most such missiles, but not by all of them. Some explode at the end of their flight even when they hit nothing. Computer-flown ships fire these missiles too, and they work the same in multiplayer, where the blast can hurt other pilots near it and every pilot sees and hears the explosion.

A weapon mod can give a gun such a warhead too: its shots explode when they pass close to an enemy ship, so a near miss still does damage, and the blast hurts the ships around it as a missile's does. A ship takes at most one blast from each shot, however many pieces of a scattered shot explode around it, and a piece that hits a ship directly counts as that ship's blast. Most such shots do not explode near friendly ships or your own, but their blasts can still catch a wingman next to an enemy. Computer-flown ships carrying such a gun open fire from a little farther away. These guns work the same in multiplayer, where the blast can hurt other pilots near it and every pilot sees the explosion.

A weapon mod can also make a gun's shot burst into a spray of smaller shots: where it explodes near a ship, where it hits one, or where its flight ends, it releases a handful of shots of the same or another gun in a cone, a ring, a fan or a straight stream, which fly on and hit whatever they meet. They are your shots, so they never hit your own ship, and the pieces a shot releases where it hits a ship never hit that ship. Computer-flown ships fire such guns too. When a fight is very crowded with shots, some of a burst's pieces may not appear. These guns work the same in multiplayer: every pilot sees the pieces fly from where the shot burst, and the pieces make no firing sound of their own.

A weapon mod can also make a gun, a missile or the torpedo hit shields and armour differently: part or all of each hit can go past the shields straight to the armour (a torpedo that ignores shields), a gun can drain shields faster or slower than its damage says, or do more or less to armour. A weapon that cannot get through a raised shield, or that never damages armour, is possible too: an ion gun that only strips shields cannot destroy a ship on its own. Every ship that carries the weapon hits this way, computer-flown ships included.

A mod gun can also be a beam weapon: while you hold the fire button it fires a continuous beam that hits whatever it points at instantly, up to its range, and keeps doing damage as long as it stays on the target. The beam uses energy while it fires and pauses when the energy runs out. A beam can overheat: held too long, it switches off for a while and then starts again. The bar above the Weapons display's `G:` line shows its heat, and while it is overheated, the time left to cool down. The lead pip sits on the target

while it is within the beam's range. Ships flown by the computer and turrets you are not sitting in do not fire beam weapons.

Some beam weapons fire in timed pulses instead: one press of the fire button starts a pulse, which keeps firing for its full time even after you let go, and then the beam rests for a while before it can fire again; holding the button fires the next pulse as soon as the rest is over. A charged beam has to be charged first: hold the fire button to charge it and release it to fire the pulse (some fire by themselves once fully charged); the longer you charged, the stronger and longer the pulse. Some charged beams also fire on a quick press: they finish charging to their minimum by themselves and fire a short, weak pulse. Releasing too early, switching views or running out of energy loses the charge, and a pulse stops early when the energy or the rounds run out, when the beam overheats or when you switch views. The bar above the Weapons display's `G:` line fills while the beam charges, empties as the pulse fires, and shows the rest in the damaged colour.

A mod gun can also overheat or fire in bursts. An overheating gun gets hotter with every shot and cools once it has rested; when it gets too hot it stops firing for a while and then starts again, cold. A burst gun fires a few shots and then pauses before it fires again; some burst guns fire the whole burst from one press of the fire button. The bar above the Weapons display's `G:` line shows the hottest such gun's heat (for a burst gun, how much of the burst is used), and while a gun is held off, the time left in the damaged colour. A mod can give the gun a sound for the moment it locks. Every ship that carries such a gun is held by it the same way, computer-flown ships included, and these guns work in multiplayer too.

A mod gun can also need ammunition. All guns of that kind on your ship share one supply of rounds, and each gun you fit adds its own magazine to it; a gun comes with a full magazine. The Weapons display's `G:` line shows the rounds left after the gun's name, `ACn(300)`, and `(0)` when they are gone; when the line is too long for the display it shortens, and with four different guns on a Centurion the count can be left off. A gun out of rounds does not fire and uses no energy, and a mod can give it a click for the moment you pull the trigger on it. Buy more at a Ship Dealer: while your ship carries the gun, the Buy list has a row for its rounds, each purchase adds a pack up to what the gun's magazines hold (NO ROOM ON SHIP when they are full), and the last purchase that fills them costs only its share. Rounds cannot be sold: selling the gun gives up the rounds it held. Computer-flown ships carrying such a gun start with a full magazine and never get more; your hired wingmen use the rounds of the ship they fly, and you restock that ship when you fly it yourself. The rounds left are kept in your saves. In a multiplayer session the server keeps count of your rounds: the Weapons display shows its count, and a gun stops firing when the server says it is empty. Each launch takes the rounds your ship carries, up to what its magazines hold, so rounds bought at a Ship Dealer during the session come with you.

A save stores a new gun as the original gun it is based on, and the `.openpriv.json` file next to the save remembers which new gun it is. A save loaded without the mod has that original gun. In Classic, or with **Weapon Mods** off, the ship carries the original gun too, and the new gun comes back when the mods are in effect again.

Weapon mods apply only in Enhanced and Modern; Classic always uses the original guns. In a multiplayer session the server's weapon mods apply instead of yours (see Multiplayer). The **Weapon Mods** box in the debug overlay's MODS section (or `WeaponMods=0` in `openprivateer.ini`) turns them off and on, and the change takes effect at once, also in flight. With no mod folder, nothing changes.

A file with a mistake is not loaded at all, and the log names the field and the reason. While any weapon mod is installed, the log also lists the guns in effect with their derived figures (shots per second,

damage and energy per second, damage per energy, range). Starting the game with `openprivateer --weapons-report` writes that list to the log even with no mod installed, and to the console when there is one; the game then starts as usual. Use plain ASCII for names: the game's fonts have no other characters. The generated `content/mods/weapons/README.txt` has the format and an example.

Ship Dealer mods (experimental)

A Ship Dealer mod changes what the Ship Dealer offers. OpenPrivateer comes with none: without one the dealer is the original's. Each mod is its own folder with a `shipstuff.json` in it, for example `content/mods/items/prices/shipstuff.json`; mods load in alphabetical order of their folder names, and a later mod's value wins.

A mod changes the dealer's existing items and can add new ones: shield generators, engine upgrades, armour, ECM packages, repair droids, the Righteous Fire improvers and Cargo Expansions of its own (new guns come from weapon mods). For each item it can set:

- its name, in the selected language when the mod gives one;
- what it costs to buy, what the dealer pays when you sell it, and what a repair costs;
- its picture in the dealer's window, for Buy and Sell and for Repair;
- the ships it is offered for, which can add a ship the original does not sell it for (for example the level 5 engine for the Centurion) or remove one;
- that it is not for sale at all;
- the bases that sell it: by base name, base type (pleasure, refinery, mining, agricultural, pirate and so on), system, quadrant, or the factions that fly in the system.

What a mod keeps off the dealer's Buy list never leaves your ship: you can always sell and repair what you carry, at any base. Trading in your ship for a new one values your equipment at the prices the dealer pays for it. An engine level bought for a ship the original does not offer it on works like any engine: it sets the ship's energy (capacity and how fast it refills), is kept in the save, and can be sold and repaired.

A mod can also change what shield generators and engines do: how much a shield holds on each side, how fast it recharges, how much energy it draws and how hard hits strike it; and how much energy an engine level holds and how fast it refills, the same for every ship or different for each type of ship. It can change how much armour each armour type gives, how often an ECM package makes a missile lose its lock, how fast a repair droid works, and how strong the Righteous Fire improvers are. These changes take effect the next time you launch, for your own ships only: ships you meet in space keep their original equipment. Damaged armour keeps its share of damage when a mod changes how much the armour holds, so repairs cost the same. Your save keeps the original equipment, so it loads normally without the mod and in Classic, where everything works as in the original.

New items. A new item appears in the dealer's Buy list with its own name, picture and prices, and you buy, sell and repair it like any item. Each one is made like a stock item (for example a "Shield Mk IV" like Shield Generator 4) and takes that item's place on your ship: a ship with the Shield Mk IV has no room for another shield generator until you sell it. It works with its own strength from your next launch, a trade-in values it at the price the dealer pays for it, and a pilot you hire flies your ship with it. Your save records the stock item it is made like, with the new item noted beside it, so the save loads without the mod and in the original game as a ship with that stock item. In Classic, or with Item Mods off, your ship carries the stock item and the dealer shows only the original items; the new item is back when the mods are in effect again. If you load a save whose new item's mod is no longer installed (or no longer makes

the item the same kind of stock item), your ship keeps the stock item and the notification center tells you which items were replaced; install the mod again and reload before saving if you want them back. In a multiplayer session the server's new items are sold, and a new item you fit works on the server as it does alone (see Multiplayer).

A Ship Dealer mod can also make the Cargo Expansion hold more. The bigger hold counts at once, everywhere you load cargo: buying at the Commodity Exchange, mission cargo, the tractor beam, buying a new ship and switching ships. Your save still records an ordinary Cargo Expansion. Without the mod (or in Classic, or with Item Mods off) the hold is the original size again: cargo already aboard stays, but you cannot load more until the hold is below the original limit.

A Ship Dealer mod can also change how many missiles or torpedoes each launcher carries, for every type of ship or differently for each, and on which positions of a ship the dealer fits a launcher. It can only take positions away, so a ship may take fewer launchers, or none: the dealer then answers NO ROOM ON SHIP. A launcher you already carry on such a position stays and fires, and rounds above the new limit stay aboard, but the dealer sells no more until you are below it. These limits count at once; your save keeps your launchers and rounds as they are, and without the mod (or in Classic, or with Item Mods off) the usual limits apply again.

Ship Dealer mods apply only in Enhanced and Modern; Classic always shows the original dealer. In a multiplayer session the server's Ship Dealer mods apply instead of yours (see Multiplayer). The **Item Mods** box in the debug overlay's MODS section (or `ItemMods=0` in `openprivateer.ini`) turns them off and on, and the change takes effect at once, also with the dealer open. Nothing a Ship Dealer mod changes is stored in your save except which new items your ships carry.

A file with a mistake is not loaded at all, and the log names the field and the reason. While any Ship Dealer mod is installed, the log also lists every item the dealer offers with its name, prices, ships and bases, and marks what each mod changed. Starting the game with `openprivateer --dealer-report` writes that list to the log even with no mod installed, and to the console when there is one; the game then starts as usual. Adding `--dealer-base=<base name>` also shows whether that base sells each item. The generated `content/mods/items/README.txt` has the format and an example.

Start mods (experimental)

A start mod changes how a NEW game starts: the base you start at (an original base or an Open Universe base such as Gemini), your credits, your standing with each faction and your kill record, guild memberships, the quadrant maps you own, the name and callsign the registration screen suggests, and your whole ship: hull, guns, launchers, missiles, equipment, scanner and cargo. Each start mod is a folder holding an `init.json`, for example `content/mods/init/mystart/init.json`:

```
{
  privateer: {
    start: "New Detroit",
    credits: 50000,
    ship: { hull: "centurion", launchers: { "2": "missile_launcher" } },
  },
}
```

The file has one part for Privateer (`privateer`) and one for Righteous Fire (`righteous_fire`); only the game you play uses its part, and anything the file leaves out stays as in the original start. Equipment uses the names the Ship Dealer shows (`"Tungsten Armor"`), guns can also use the short weapon names (`laser`), and guns and launchers are placed on numbered hardpoints. The hull can also be a ship mod's ship you fly (`hull: "capital/captain"` , see "Ship mods"); that ship mod must be installed, otherwise the start mod is not used and the log says why. Equipment can also be a Ship Dealer mod's new item by its name in the mod (`shield: "shipyard/mk4_shield"` , or `gun_cooler: "shipyard/chiller"` in place of `true`); that Ship Dealer mod must be installed in the same way.

Only one start mod can be used. With a single folder it is used automatically; with several, pick one in the launcher's Mods section (**New Game Mod**), in the debug overlay's MODS section (**Init Mod**) or with `InitMod=<folder>` in `openprivateer.ini` , otherwise none is used and the log lists them. The **Init Mods** box (or `InitMods=0`) disables start mods. They apply only in Enhanced and Modern and never in a multiplayer session; Classic always starts as the original does, and Righteous Fire's "Load an old Privateer ship?" ignores them. A start at an Open Universe base needs Open Universe on; otherwise you start at the original starting base.

The main story may not work when a start mod changes the start: the story expects the original start.

A new game from a start mod saves like any other game, so the save loads in Classic too. A file with a mistake is not used at all, and the log names the field and the reason. The generated `content/mods/init/README.txt` has the format and an example.

Mission mods (experimental)

A mission mod adds missions to the mission boards: the Mission Computer terminal and the Mercenaries' and Merchants' Guild computers offer them beside their usual random missions. Each mod is a folder holding a `missions.json` , for example `content/mods/missions/mymod/missions.json` :

```
{
  missions: {
    troy_sweep: {
      kind: 'patrol',
      board_text: 'Sweep $NN nav points in $DS for $EN raiders. $PY credits.',
      summary: 'Patrol $NN nav points in the $DS system.\n\n$CL pays $PY credits.',
      destination: { systems: ['Troy'] },
      offered: { boards: ['terminal'], weight: 100 },
    },
  },
}
```

A mod mission is one of the usual kinds: patrol, scout, defend, attack, bounty or cargo. You take it, fly it and get paid exactly like any other board mission: it fills one of your three mission slots, shows its objectives on the Navigation Map and in the cockpit, pays when you land after completing it, and appears in the notification center. What the mod chooses is the text the board shows and the text your Mission Computer keeps, who pays, where the mission sends you, which enemy you meet and how many, the cargo, the pay, and at which bases and on which boards it is offered. Words starting with `$` in the texts are filled in when the board shows the mission (`$DS` the system, `$PY` the pay and so on; the generated README lists them all). A mission without `destination.systems` goes where the usual missions go: most often the system you are in, otherwise a system picked the original game's way, which favours the Clarke quadrant wherever you are; the generated README explains the rule.

Each board slot picks between the usual missions and the mod missions offered there, so a board shows a mix. At an Open Universe base such as Gemini the boards offer the usual missions of the original base it borrows its rooms from (when it names one) and the mod missions offered there; without mission mods they stay empty. Mod missions can also send you into an Open Universe system such as Gemini itself (a patrol of its nav points, an attack at one of them) and deliver cargo to, or defend, one of its bases or stations: for a delivery, land at that very base.

Mission mods apply only in Enhanced and Modern, and never in a multiplayer session; Classic always shows the original boards. The **Mission Mods** box in the debug overlay's MODS section (or `MissionMods=0` in `openprivateer.ini`) turns them off and on; the change applies the next time a board shows its missions, and a mod mission still on a board can no longer be taken once they are off. Missions you already carry keep going: your save stores the whole mission, so it flies on without the mod and in Classic too. The one exception is a mission into an Open Universe system or to an Open Universe base: it is kept in the save's `.openpriv.json` file and is there only when you load the save in Enhanced or Modern with Open Universe on and the same Open Universe content installed. Otherwise the mission is gone from your list, and the notification center tells you which one and why.

A file with a mistake is not loaded at all, and the log names the field and the reason. Starting the game with `openprivateer --missions-report` writes the list of mod missions to the log and the console; the game then starts as usual. The generated `content/mods/missions/README.txt` has the format and an example.

Ship mods (experimental)

A ship mod adds variants of the computer-flown ships: a ship based on a stock one, such as the Paradigm, with its own name on the target display, its own shields and armour, other guns, missiles and turrets, its own speed, turning and energy, its own loot, and its own effect on your standing with the factions when you destroy it. A ship mod can also bring a new ship with a look of its own: its own pictures in flight and on the target display, its own size, and its own places where its guns sit and where hits land. A mod ship can carry up to 24 turrets, each covering its own side of the ship, and a mod ship can be a **station**: it never moves or turns, but its turrets and guns fire at you as any ship's do, and you destroy it as any ship. An Open Universe mod places these ships in its encounters, and can make a station a carrier you land on (see the universe mods above). Each mod is a folder holding a `ships.json`, for example `content/mods/ships/capital/ships.json`. A variant can have a **phase shield**: while that side of its shield is up, your guns, missiles and torpedoes do nothing to it, the hits land and the shield takes them without draining. Only a weapon a weapon mod makes able to go through a phase shield, such as an anti-matter gun, reaches it; once such hits drain that side or wreck its shield generator, every weapon gets through there. On the Target MFD, a side whose phase shield is up has a thin light blue layer just outside its shield bars; the bars themselves show and drop away as on any ship, and a side without the blue layer is an ordinary shield. Some variants play their own sound when their phase shield stops a hit. You never buy a variant, and a ship of it you meet is never in your saves.

A ship mod can also bring a ship **you fly**, a mod ship with its own speed, turning, size, armour, damage areas, gun positions and pictures in the outside views (F7 to F9); your ship looks like itself in the autopilot flyby and the jump sequence too. Each such ship is based on one of the four ships you can own, its "alias": it uses that ship's turrets, Weapons display and death pictures, and your save records it as that ship. The takeoff and landing movies show the mod ship itself, flying the alias's path past the base (a mod can keep the alias's ship there instead). It can take a moment to appear when the movie starts right after you first fly the ship. A mod can give it a cockpit of its own, a repaint of the alias's: the views, gauges and damage marks then have the mod's pictures (a view the mod leaves empty shows open space with only the HUD), its windows, gauges and displays where the mod puts them, and its energy and fuel gauges go dark the way the mod says (a cockpit taken from another Wing Commander game drains them as that game does), while the turrets are the alias's. The drained part of a gauge shows black, or a picture the mod gives it: a cockpit taken from Wing Commander 1 or 2 shows the game's dim "empty" cells there (the Sabre's fuel tank empties into dark blue cells). Such a cockpit can also have its own MFDs, for example two displays on a ship whose alias has one: each shows the pages the mod gives it, `[` cycles the left one and `]` the right one, and the page keys bring each page to its display (a cockpit taken from Wing Commander 1 or 2 shows the weapons and damage pages on the left display and the destination, target and comm pages on the right, as that game does). A cockpit can also show your shield strength as numbers in the front view, which count down as that side of the shield takes hits and climb back as it recharges (a cockpit taken from Wing Commander 1 or 2 prints the front and rear shield beside its shield display, as that game does). The number is a percent of that side's full shield: 100 at full, 1 to 99 as it takes hits, and 0 only when that side is empty or the shields are switched off or none is fitted; a damaged Shield Generator or a lowered shield power level keeps it below 100. A

mod can show the Shield Generator's strength itself instead. A cockpit's turret view (F5) can have its own radar, centred on where the turret points, and can leave out the turret MFD. A cockpit taken from Wing Commander 2 shows its gunner's seat in the turret view, with the gun energy column as the turret's energy gauge and the gunner's scanner as the radar; that game's seat is its rear view, so F4 there looks out behind the ship with no cockpit around it. A mod ship without a cockpit of its own starts each flight in the cockpitless HUD (see section 14), and `,` shows its alias's cockpit. You buy one at a Ship Dealer that sells it (click the salesman or press `B` in the showroom, see section 11), at the price the mod gives it, and it trades in at that price or at the share of it the mod sets; or you start a new pilot in one with a start mod (see "Start mods": `ship: { hull: 'capital/captain' }`). Fitted, parked and flown by a hired wingman, it works like any ship you own. Its own gun and launcher positions and its hold (at most its alias's) are what you fit and carry; the Ship Dealer offers it the equipment the mod allows, and Repair & Upgrade and the Launch Pad show the mod's picture of it, or its alias's; a mod can give some launch pads a picture of their own, seen from that pad's camera.

Your saves keep the mod ship as long as the mod is installed. Loaded without the mod, the ship is its alias: guns and launchers that had no place on the alias are not on it, the log says so and the notification center shows a message; save over it only when you no longer want the mod ship back. In Classic and with **Ship Mods** off the ship flies as its alias and stays the mod ship in your save. In a multiplayer session you fly a mod ship when the server runs its mod (see "Flying a mod ship in multiplayer" in the multiplayer section); otherwise you do not join, and the game continues in single player.

A ship mod can also change a stock ship itself: then every computer-flown ship of that kind, for example every Centurion you meet in the encounters and in the missions, has the mod's armour, guns, speed or name, and can have new pictures, a new size and new places where its guns sit and where hits land. Your own ship stays as it is and keeps its own look in the outside views, and so does an owned ship a hired pilot flies for you or that comes along as a wingman. Without the mod, every ship is the stock one again.

A ship mod can also change the Tarsus, Orion, Centurion and Galaxy you fly: their speed, acceleration and turn rates, their size and where hits land, where their guns sit and which positions you can fit, their cargo hold, their armour when you have none fitted, and what the Ship Dealer asks for them, what they trade in for and what they come with. Clicking the Centurion in the showroom then buys the changed Centurion, and it flies that way from your next launch; the equipment you buy is still the Ship Dealer's, and your ship keeps its own look in the outside views. Your saves record the stock hull, so without the mod, in Classic and with **Ship Mods** off the ship flies as the stock one. In a multiplayer session the server's ship mods decide: when they change a hull you fly, it flies changed for you and every other pilot, otherwise it flies as the stock one.

Ship mods apply only in Enhanced and Modern; Classic never shows a variant. In a multiplayer session the server's ship mods apply instead of yours (see Multiplayer): the changed stock ships are changed for everyone, and new mod ships appear where the server's universe mods place them in the original Gemini systems. The **Ship Mods** box in the debug overlay's MODS section (or `ShipMods=0` in `openprivateer.ini`) turns them off and on; the change applies when you next enter a star system, and with them off the encounters that would bring a variant bring none and every ship is the stock one. Starting the game with `openprivateer --ships-report` writes the list of variants and changed stock ships to the log and the console; `openprivateer --ships-preview <folder>` writes each mod ship's and each changed stock ship's pictures, in the game's own colours, into that folder.

While the ship mods load, the loading screen counts them ("Loading ship mods: wc3ships 12 of 36"). The game reads a mod ship's pictures the first time that ship appears in flight, so starting the game stays quick however many mod ships are installed. If one of its pictures is damaged, the log says so then and the ship shows the pictures of the ship it is based on; `--ships-report` checks every picture at once. The generated `content/mods/ships/README.txt` has the format and an example.

To make a new ship from a stock one, start from the asset export (section 25): every ship that flies in the game's encounters has a `ship_mod.json5` in its folder there, the ship with all its values as a ship mod entry. Copy that folder into your mod (for example `content/mods/ships/capital/GOTHRI/`), name it in the mod's `ships.json` as `{ ships: [ { file: 'GOTHRI/ship_mod.json5' } ] }`, and the mod has a ship that is exactly the stock Gothri, with its own pictures from that folder. Give it a new `key`, change the values and repaint the pictures you want to change, and place it with an encounters row. To change every Gothri you meet instead, name the same folder as a changed stock ship, `{ override: { GOTHTYPE: { file: 'GOTHRI/ship_mod.json5' } } }`: as exported it changes nothing, and what you change or repaint there changes every computer-flown Gothri. The Talon's two folders can be used this way only without their pictures (delete their `art`).

The folders of the four ships you can fly (`tarsus/`, `orion/`, `centurion/`, `galaxy/`) make a ship you can fly as well: copied into a mod as they are, the Ship Dealer sells the copy at the stock ship's price, and its Weapons MFD shows the stock page, drawn from `mfd_weapons.png` in the folder with each gun and launcher icon at its `mfd` point in `ship_mod.json5`. Repaint that picture and move the points for a ship of your own. To use one of these four folders for a computer-flown ship only, or to change every computer-flown ship of that kind, delete `player`, `art.mfd_weapons` and every `mfd` from its `ship_mod.json5` first.

OpenPrivateer also ships built-in Open Universe content. In Enhanced and Modern with Open Universe enabled, **Gemini** is a flyable system in Humboldt, the metropolis of the sector. Its nav points:

Nav	What is there
1	**Gemini**, the metropolis planet (landing base)
2	**Heliopolis Array**, a solar array station (no weapons fire)
3	**Novaya Riga**, a run-down refinery station
4, 5	Asteroid fields
6, 7	Empty space
8, 9, 10	Jump points to Troy, Junction and Aquila

Return jumps connect those systems back to Gemini: Troy at Nav 9, Junction at Nav 10 and Aquila at Nav 1. You can fly through Gemini with autopilot and leave through any of its three jump points. On the Navigation Map, QUAD and `Home` show the quadrant of the system you are in (Humboldt for Gemini, Petrov for Aquila). Selecting one of Gemini's nav points shows "Jump to <system>" for a jump point, the station's name and type for a station, and "Asteroid Field" or "Empty Space" for the plain nav points. Gemini's sky shows no sun or planets yet.

Gemini has traffic like an original system. Near each nav point you may meet militia patrols in Gladius fighters, merchants in Tarsus and Drayman transports, or pirates in Talons and in Demon fighters; around the planet a militia Paradigm sometimes patrols. Pirates are most common around Novaya Riga and the asteroid fields, the militia around the planet and the stations. Which group you meet at a nav

point is decided each time you enter the system, and its ships appear when you come near the point. Autopilot stops when its route crosses another nav point's area, as in an original system.

Weapons cannot be fired near the **Heliopolis Array**: within about 12000 meters of it your guns, turrets and missiles stay silent and the message line reads "No weapons fire zone". Other ships are not bound by it. The tractor beam still works.

A visit to a built-in mod base uses the interior of a similar original base, but the mod base keeps its own name and location and only its configured services are available. Its Mission Computer and guilds have no missions yet, and missions cannot be delivered there. You launch back into Gemini. Saving while docked at a mod base stores the extra location in the matching `.openpriv.json` sidecar while the original save stays in the original format. If you turn the built-in content or Open Universe off before loading, you load at the similar original base instead.

The asteroid fields at Nav 4 and Nav 5 work like original asteroid fields: rocks can hit the ship (a rock bursts on impact, so each one hits only once), and autopilot stops when it reaches a field.

A `navpoints.json`, `zones.json` or `encounters.json` that reuses a built-in key with the same `mod` and `key`, or adds a second jump between the same systems, is not loaded. An encounter table on a built-in nav point or zone is added beside the built-in one.

To disable the built-in content, uncheck **Built-in Content** under **Open Universe** in the debug overlay, or set `UniverseBuiltinContent=0` in `openprivateer.ini`, then restart.

Translations live under `content/i18n/` with names such as:

```
openprivateer_de.i18n
```

The translation format uses `key=value`, `#` comments and `\n` for embedded line breaks.

The original fonts have limited character coverage. Plain ASCII or transliteration is safest unless a suitable custom font is used.

21. Optional 1993 Alpha Content

The `alpha/` folder is for optional material from the August 1993 Privateer alpha.

It is not distributed with OpenPrivateer and is not required to play.

See `alpha/README.txt` for the exact files expected by your build.

22. Multiplayer

OpenPrivateer includes dedicated client/server multiplayer. It is still an Alpha feature.

Server

Basic example:

```
openprivateer_server --tre /path/to/PRIV.TRE
```

The default UDP port is `24601`.

Useful server options include `--port`, `--max-players`, `--verbose` and `--experience-mode`.

`--all-equipment /path/to/RF.TRE` makes a Privateer server's sessions sell and fly Righteous Fire's equipment (see section 11). It needs `--tre` and is refused in Classic and with Righteous Fire data. Every player must set **All Equipment** the same way as the server: a client that differs is refused with "All Equipment mismatch". On a server without it, equipment from Righteous Fire on a pilot's ship (the Fusion Cannon, Shield Generator 6 and 7, Isometal Armor, the Gun Cooler) is not used in the session; the save keeps it.

A server can run weapon, Ship Dealer, ship and universe mods: `--mods DIR` takes a folder laid out like your own `content/mods` (for example `DIR/weapons/<mod>/weapons.json`, `DIR/items/<mod>/shipstuff.json`, `DIR/ships/<mod>/ships.json`, `DIR/universe/<mod>/encounters.json`, or a pack in `DIR/packs/<mod>/`), with an optional `DIR/mods.json` for the order. Only weapon, Ship Dealer, ship and universe mods run in multiplayer; a server with Ship Dealer mods gives every pilot the modded shields, armour, engines, improvers and new items, each ship's own where a mod sets them per ship, and a ship mod that changes a stock ship changes every computer-flown ship of that kind in the session, a ship mod that changes a stock ship you fly changes it for every pilot in the session, and the mod ships a ship mod lets you fly can be flown there (see "Flying a mod ship in multiplayer" below). Of a universe mod, a session runs only the encounter tables in the original Gemini systems: they add, take away or change the original's groups of ships at a nav point, exactly as in single-player, and can bring a ship mod's new ships into the session; its new systems, nav points and zones, and its tables in those systems, are not used in multiplayer. The server does not start with `--mods` in Classic, with a mod file that has a mistake, or with a weapon a session cannot run yet: a charge or beam (hitscan) gun, a gun or missile with lethal hits, a phase torpedo's decode or a missile's timed lock; the message names the mod and its file. `--mods-report` lists each mod with its content hash.

Client

To join from the launcher, open its **Multiplayer** section, right below **Experience**:

- **Server**: the server's address, `HOST` or `HOST:PORT` (for example `192.168.1.20` or `play.example.org:7000`). Without a port, `24601` is used. IPv6 addresses are not supported.
- **Callsign**: your name in multiplayer, up to 24 characters (letters, digits, spaces and common punctuation). Empty means `Pilot`.
- **Check Server** asks the server which mods it runs, without joining, and compares them with yours. It shows whether the server answered (or why not), its version and mods, and whether all of its mods are installed on this computer, or which are missing or different.

With a server entered, the section title reads "PLAY joins" and the address. **PLAY** then checks the server first ("Checking the server…"; it takes at most a few seconds) and starts the game only when the server answers and all of its mods are installed on this computer; the session is joined once you load a game or start a new one. A successful **Check Server** in the last 30 seconds counts, so PLAY starts at once after it. If the server cannot be reached, answers with another version, or runs mods this computer lacks or has in a different version, the game does not start: the Multiplayer section opens with the reason and

“Invalid server address. Correct it or leave the Server field empty to play single-player.” Start the server, fix the address or install the server’s mods and press PLAY again, or clear **Server** to play single-player. Both fields are remembered for the next start. Multiplayer uses the server’s weapon, item (Ship Dealer), ship and universe mods instead of your local mods.

From the command line:

```
openprivateer --connect HOST:24601 /path/to/PRIV.TRE
```

`--connect` skips the launcher, and `--callsign NAME` sets the callsign; both take precedence over the launcher’s fields.

The server can reject clients when important session properties differ, including build, Privateer vs. Righteous Fire data, Experience mode or All Equipment. When the game is started to join a server, the All Equipment box is locked for that run: set it before starting.

When joining, the game first asks the server which mods it runs, during the loading screen. If this computer has all of them (the same mod folders with the same files, a ship mod’s pictures included; your own load order and on/off choices do not matter), the game loads exactly the server’s weapon, Ship Dealer and ship mods for the session, so everyone flies the same weapons and equipment, sees the same dealer and meets the same changed ships; the server’s universe mods decide which ships meet you at the original nav points. A universe mod counts with all of its files, also the ones a session does not use. Your other mods (missions, new game, and your own universe content) stay off in a session.

Flying a mod ship in multiplayer. A mod ship you fly (section 11) joins a session only when the server runs its ship mod. It then flies exactly as in single-player: its own speed, turning, size, armour and gun positions, and the other pilots see it with the mod’s pictures. If your ship comes from a ship mod the server does not run, the game does not join (the log says “join refused: the server’s session does not run the ships mod …”), and you keep playing single-player; fly one of your stock ships or one of the server’s mod ships to join. A ship of a mod the server does not run, chosen after joining, does not launch. At a Ship Dealer, the salesman or `B` in the showroom lists the stock ships and the server’s mod ships that dealer sells, bought as in single-player. If you buy one whose mod is not among your own installed and enabled mods, the showroom’s status line says so: your save keeps the ship, but outside the session it loads as its alias until you install and enable that mod.

You fly a session with at most as many missiles and torpedoes as the server’s dealer lets your launchers carry; your save keeps your own count. Your afterburner, Speed Enhancer and Thrust Enhancer work as in single-player: without an afterburner the afterburner key gives no boost.

If the server runs a mod this computer does not have, or a different version of it, the game does not join: the title screen shows “Join refused. Server mods differ.” (click to dismiss), the log names each missing or different mod with its size, and you can play single-player with your own mods. Install the server’s mods (for example by copying its pack folder into `content/mods/packs/`) and start again to join; **Check Server** in the launcher shows the same list before you start. Mods are never downloaded from a server.

If you have the same mod files but your copy of the game cannot load one of them (usually because the server runs a newer build that knows fields yours does not), the game does not join either: the title screen shows “Join refused, weapons mod” (or “items mod”, “ships mod”, “universe mod”) with the

mod's name and "did not load.", and the log names the mod and the exact error. You can play single-player, but your weapon, dealer-item, ship and mission mods stay off until you restart the game. Use the same build of the game as the server (or one the server's mods were made for) and start again to join. A mod that loads with warnings joins normally.

When a session ends (you leave it or your ship is destroyed), your own weapon, dealer-item, ship and mission mods stay off until you restart the game; the title screen says "Restart to use your own mods.".

Files in the `content/priv` or `content/rf` override folders only change your own game: the server uses its own copy of the game data. The log warns about them when you join a server.

Multiplayer covers the shared flight/combat environment. The normal single-player campaign progression is not the multiplayer campaign model.

For now, in a multiplayer session the Mission Computer and the guild boards offer only patrol, scout and cargo missions. They are tracked on your own computer and paid as usual. Attack, defend and bounty missions (and a scout whose nav point is a base, which also has to be cleared of hostiles) need ships that are not yet part of the shared session, so they are not offered there. Such a mission you already carry waits: its ships do not appear in the session, and it stays on your list for when you play single-player. A board that has nothing it can offer says it has no missions at this time.

23. Update Check

Release builds can check for a newer OpenPrivateer build at most once every 24 hours.

Nothing is downloaded or installed automatically.

When a newer build exists, the title screen shows:

```
New build available

Click to download
```

Left-click opens the download page. Right-click dismisses that particular build notice.

The launcher shows the same news as an **Update available** line with a **Download** button. On Windows and Linux, Download opens your browser on the matching archive of the newest build; on other systems it opens the release page, which lists every archive. Unpack the archive yourself to update.

The check can be disabled in settings or for one launch with:

```
--no-update-check
```

24. Original Developer Switches

OpenPrivateer supports several switches from the original game:

```
-K
```

Skip the intro.

```
-B<n>
```

Start at original base ID `n`.

```
-D<n>
```

Set starting credits to `n`.

Example:

```
openprivateer.exe -K -B39 -D100000
```

These are mainly useful for testing and development.

The launcher's **Original Developer Switches** section sets the same three as checkboxes that are remembered and apply on every start. Switches given on the command line win. The `-B` and `-D` number fields can only be edited while their box is ticked.

Other command-line options

```
openprivateer --help
```

lists every option. The ones for files and the launcher:

Option	Effect
`--home DIR`	Use `DIR` as the user-data folder for this run
`--portable`	Use the program's folder as the user-data folder
`--data DIR`	Look for the game files in `DIR` first
`--launcher`	Show the launcher
`--no-launcher`	Never show it; exit with an error if files are missing

`OPENPRIV_HOME` and `OPENPRIV_DATA` (a list separated like `PATH`) are the environment-variable versions of `--home` and `--data`.

`--gpu-debug` (or `OPENPRIV_GPU_DEBUG=1`) runs the graphics device in debug mode for troubleshooting; see "The game closes or crashes while loading" in Troubleshooting.

25. Exporting the Game's Assets

```
openprivateer --export-assets C:\PrivateerExport
```

writes the pictures, sounds, speech, palettes, text and data of your own `PRIV.TRE` and `RF.TRE` into the folder you name, and exits. No window opens; the progress is printed in the terminal. The game files are

found the same way the game finds them (the launcher's Game Data folders, `--data`, `OPENPRIV_DATA`, a GOG `GAME.GOG`); `--tre FILE` and `--rf FILE` name an archive, or a `GAME.GOG`, directly. The export always reads the archives themselves: files in `content/` do not change it.

The folder holds `priv/` and `rf/`, laid out like the archives, with each archive file as a folder of its parts. A container's parts are numbered (`000`, `001`, …), nested containers as subfolders:

File	Content
`.png`	a picture, in the colours the game shows it
`_sheet.png`	every frame of a set of pictures side by side
`_sheet.json`	where each frame of the sheet sits
`.mask.png`	a shape the game fills with a colour (white)
`hud.json`	a cockpit's gauge and screen positions
`faces.json`	which portrait parts make up each character
`.wav`	sound effects and speech, the original samples
`.swatch.png`	a palette, 16 colours per row, index 0 top left
`.gpl`	the same palette for GIMP and other paint programs
`.palette.json`	the same palette's colours as text
`chunks.json`	a data file's structure, each part's bytes as hex
`.txt`	captions and other text
`.json5`	a mod file to copy and edit (see below)
`.bin`	everything else, unchanged (pictures among them)

The pictures written are: every ship from all 37 directions with its afterburner flames and its targeting and weapons display art, the planets, bases, suns, explosions, missiles and other space objects, the stars, the four cockpits (each view as a 320x200 screen, the gauges, crosshair and damage marks), the radar icons, the pause plaques and their lettering, the in-flight comm faces, the conversation portraits and their backgrounds, the title, Options, Mission Computer, joystick calibration and mouse cursor art, the jump, takeoff and landing movie art, every picture of the base screens (`OPTSHPS.PAK`, and Righteous Fire's `OPTSHPS2.PAK`: the backdrops, the people, the animations, the shop and guild widgets), the Ship Dealer's and Software Booth's item icons (`SHIPSTUF.IFF`, `gunt/` and `soft/`, a broken item's Repair picture as `_broken`), the Software Booth's overlays and the Commodity Exchange's icons, named after the goods. The fonts and the other movies are still `.bin`.

A base picture is coloured with the palette of the screen that shows it. A few are shown on more than one screen in different colours (a figure that stands in several bars): they are written in the first screen's colours, and `--all-palettes` also writes them in each of the others, as `000.palt22.png` for palette 22. A few pictures appear on no screen at all; they get a palette that is a guess, marked `guessed` in the manifest, and the export prints a note naming them.

`DATA/OPTIONS/OPTIONS.IFF/scenes/` holds one JSON file per base screen: its palette, every picture it is built from with its place on the screen (the picture's hotspot goes there), its animated figures and their

click areas, and its click areas with their labels (“Bar”, “Launch”, …). Each entry names the exported picture files, so a screen can be put back together from them.

A picture’s frames that belong together (a ship’s views, an animation, a portrait’s eye and mouth frames) are all the same size, so they line up when stacked; a ship’s views are named by the direction they show it from (`f15_e+00_a000.png` : level with the ship, from straight ahead). The game draws the views from 30 to 150 degrees mirrored for the other side of the ship. Pictures are indexed PNGs that keep the game’s own colour numbers, which paint programs show like any other PNG; a few translucent ones (the jump point, the tractor beam’s glow) are see-through as the game blends them, which only approximates how they look over a starfield.

`priv/ships/` and `rf/ships/` hold one folder per ship, everything a mod maker needs about it in one place. The four ships you can fly are `tarsus` , `orion` , `centurion` and `galaxy` ; every other ship is named after its picture file (`GOTHRI` , `TALPIR` , `SALTHI` , …). Each folder has the ship’s 37 views and afterburner flames (`hull/` , `burn/`), its targeting display pictures (`mfd_target/`) and `ship.json` : the ship’s size, collision boxes, gun and launcher mounts, flight values and armour, and for every ship type the game flies it as, its guns, launchers, missiles, turrets, shields, energy and cargo, who flies it and the values the game works out from them, named as the mod files name them (`laser` , `missile_launcher` , `heat_seeker` , `Tail Turret`). The game does not read `ship.json` ; it describes the ship.

Every ship that flies in the game’s encounters also has `ship_mod.json5` : the same ship as a ship mod entry, every value as the game has it and its pictures taken from the folders beside it. Copied with its folder into a ship mod and named in that mod’s `ships.json` (`{ ships: [ { file: 'GOTHRI/ship_mod.json5' } ] }`), it makes a ship exactly like the stock one, to change from there (see “Ship mods”); the four ships you can fly are flyable copies too, sold at the Ship Dealer. `ships.json` in `priv/ships/` and `rf/ships/` names all of them, so the whole folder copied as one ship mod brings a copy of every ship.

The folders of the four ships you can fly also hold:

- `death/` , `takeoff/` and `landing/` : the ship’s explosion and its takeoff and landing animation;
- `cockpit/` : the cockpit views, gauges, `hud.json` and the damage marks (`decals/`);
- `repair/` : the Repair & Upgrade picture of the ship (`ship.png` , on the 320x200 screen) and every gun, launcher, engine and scanner picture of that screen cut out on its own (`hardpoint_2_laser.png` is a Laser on hardpoint 2), with `hardpoints.json` giving where each one sits on the screen and where the Ship Dealer’s click areas for each mount and turret are. A gun picture with its `x` and `y` is exactly what a weapons mod’s hardpoint image takes, and the `weapons_json` part of `hardpoints.json` is a `weapons.json` that puts every gun picture back where it was: copy it and the `repair/` pictures into a weapons mod and repaint the ones you want;
- `dealer.json` : the ship’s price, the Ship Dealer items offered for it and those left out, its engine and shield levels, turrets and mount positions; its `shipstuff_json` part is a dealer-items mod that keeps all of that as it is, to start a mod that changes it.

`priv/weapons/` and `rf/weapons/` hold the guns and missiles for weapon mods (`weapons.json` , section 20): one folder per gun with its bolt animation (`bolt/`), its Weapons display icons (`mfd_forward.png` , `mfd_turret.png`), its Ship Dealer pictures (`shop.png` , `shop_broken.png`) and its firing sounds (`fire.wav` , `fire_far.wav` for the same shot from another ship); `guns.json` and `ordnance.json` give every gun’s, missile’s and launcher’s values in the mod file’s own names and units; `stock_weapons.json5` is a `weapons.json` that states every one of those values as the game has it.

Copied unchanged to `content/mods/weapons/<name>/weapons.json` it changes nothing: change the numbers you want and delete the rest.

`priv/dealer/` and `rf/dealer/` do the same for the Ship Dealer: `icons/` holds every item's picture and Repair picture, `rows.json` every item's prices, the ships it is sold for and what it does, and `stock_shipstuff.json5` is a dealer-items mod (`shipstuff.json`) stating all of it, the pictures included. Copy it with the `icons/` folder to `content/mods/items/<name>/` as `shipstuff.json` and it changes nothing until you edit it.

`priv/interiors/` and `rf/interiors/` hold every base's interior for interior mods (section 20): `interiors.json5` has one interior per base (`stock-<number>`) with every screen of the base (the landing pad, the concourse, the bar, the Ship Dealer and the rest) as its own picture, with its animations, its figures, its bartender, the Ship Dealer's purchases and its doors; the guild offices, the Mission Computer, the Commodity Exchange, Repair & Upgrade and the Software Booth are the game's own screens behind its doors. `art/` holds the pictures, `stock_roles.json` every base's screens and their click labels. Copied as `interiors.json` into an interior mod with its `art/` folder, a base or station naming one with `interior` and its original base with `like` gets that base's screens, without its story characters (as at every added base); change the pictures and doors you want from there.

`palettes/index.json` lists every palette of a game. `INITCFG.PAK` and `INITRF.PAK` are copied unchanged when they are found next to their archive. `manifest.csv` (and `manifest.json`) lists every file: the archive file and part it came from, how it was made, and where a replacement goes. A picture's row gives its size, its hotspot (the pixel the game places on the object's position) and the palette that coloured it; `palette_confidence` `engine-rule` means that is the palette the game draws it with, `guessed` that no screen shows the picture and `palette_rule` says how its palette was chosen, `none` marks a mask; `palette_alternatives` lists the other palettes of a picture several screens show. A sound effect's row names the weapons-mod field that takes a `.wav` like it, a Weapons display icon's or a Repair & Upgrade gun picture's row the field that takes its picture, and `dealer.json` 's row the dealer-items mod fields it restates.

Options:

- `--only LIST` exports only these kinds, comma separated: `images` , `cockpits` (the cockpit pictures and `hud.json`), `ships` (the `ships/` folders), `weapons` and `dealer` (the `weapons/` and `dealer/` folders), `interiors` (the `interiors/` folder: every original base's interior as an interior mod, with its pictures), `scenes` (the base screens' JSON), `data` (here `faces.json`), `sounds` , `speech` , `palettes` , `iff` (the data files' structure), `text` and `raw` (the `.bin` files); `all` (the default) and `modkit` (what mod makers need, the `ships/` , `weapons/` , `dealer/` and `interiors/` folders among it) are presets.
- `--game priv` or `--game rf` exports one game only (default: both).
- `--raw` also writes every archive file unchanged into `priv-raw/` and `rf-raw/` .
- `--rgba` writes every PNG as RGBA instead of indexed.
- `--all-palettes` also writes a base picture several screens show in different colours in each of them.
- `--overwrite` replaces an earlier export in the folder.
- `--verbose` prints one line per archive file.
- `--home` , `--data` and `--portable` work as for a normal start.

`--only` also accepts `music`, `fonts` and `movies`; this version writes no files for them.

The files in `priv-raw/` and `rf-raw/` sit at the same paths `content/priv/` and `content/rf/` use (section 20), so a file you change there can be copied straight into `content/` to replace the original.

The folder must be new or empty. Running the export again into the same folder needs `--overwrite`; the earlier export is replaced only once the new one is complete, and other files you put in the folder are left alone. The export refuses the folder of the game files, the program folder, the user-data folder itself (a folder inside it is fine) and anything inside `content/`. Ctrl+C stops the export; what was written stays in `priv.partial/` or `rf.partial/`, and the next run with `--overwrite` removes it.

Each game is written into `priv.partial/` (or `rf.partial/`) and renamed to `priv/` when it is complete. Another program that has the folder open (a virus scanner, an Explorer window, an editor watching the folder, a terminal inside it) can stop that rename; the export keeps trying for up to 30 seconds and says it is waiting. If the folder still cannot be renamed, it stays `priv.partial/`, which is complete, and the export goes on with the next game. The summary at the end lists such folders: close what has them open and rename them by hand (`priv.partial` to `priv`), or run the export again with `--overwrite`.

Tip: export into a plain folder of your own, not into a source code or build folder or the program folder. Editors, indexers and virus scanners often watch those and hold the new files; the export prints a note when the folder lies inside one.

`openprivateer --export-assets --help` lists the options. The exit code is 0 when the export is done, 1 for a mistake in the options, 2 when a game file is missing or cannot be read, 3 when the folder is refused, 4 when a file cannot be written, 5 when everything was written but a folder had to stay `.partial`, and 130 after Ctrl+C.

The exported files are made from your own game and are copyrighted: keep them for your own use and do not share or upload them.

26. Debug Tools

The Alpha contains debugging tools used during reconstruction and testing.

Open the settings/debug overlay with:

```
~
```

or the key chosen with `DebugOverlayKey=` (see section 5).

The settings on the left are grouped into sections, top to bottom: GENERAL, GAME, VOLUME, SUBTITLES, GRAPHICS, CONVERSATION PLAYER, MODS, EXPERIMENTAL and UPDATES. They start folded: click a section's header to open it, and again to fold it away; the game remembers this until you quit. MODS holds everything about mods: what this run did with each mod (folded under its own **This run** header, which opens the same way), the reload buttons, Auto reload, the mod manager button, the switches for each kind of mod, Init Mod and Hull Atlas. EXPERIMENTAL holds the testing switches. The right-hand pane is the Mods log (see section 20).

Toggle the base click-region overlay with:

```
R
```

The region overlay draws clickable areas in green and is useful for screenshots when reporting a base UI problem.

27. Troubleshooting

NEW is greyed out

The matching INIT file is missing. Privateer needs `INITCFG.PAK`; Righteous Fire needs `INITRF.PAK`. Copy it in with the launcher (hold Shift while starting if it is turned off), or put it next to its `.TRE` file.

The game data cannot be found

The launcher lists every folder it searched and which files it found. Use **Import to Data Folder...** to copy files in, or **Add Folder...** to use them where they are. On Linux, a file named `priv.tre` is reported as "wrong letter case": rename it to `PRIV.TRE`.

With the GOG release, `PRIV.TRE` and `RF.TRE` are inside `GAME.GOG`: use **Extract from GOG...** under Game Data (section 1, "GOG installation"). If the launcher says the file is not a disc image it can read, is incomplete or damaged, check that you selected `GAME.GOG` from the GOG installation folder, and reinstall or re-download it if the message stays.

The asset export stops with "no PRIV.TRE or RF.TRE found"

The export lists the folders it searched. Start the game once and add your game folder in the launcher's Game Data section, or pass the archive with `--tre FILE` (and `--rf FILE`) (section 25).

Where are my settings and saves?

Open the launcher's **Files and Diagnostics** section. It shows the user-data folder currently in use and provides buttons for the folder, savegames, logs, music, wallpaper and content.

An old save does not appear

Privateer saves must be `.SAV`, Righteous Fire saves `.PRS`. Put them in the `savegame/` folder of the user-data folder.

"libSDL3.so.0: cannot open shared object file"

On Linux this means the program found neither the bundled SDL3 nor a system SDL3. Usually the archive was not extracted completely: extract it again so the `libSDL3.so*` files sit next to `openprivateer`, or install SDL3 3.2.16 or newer from your distribution.

The controller is detected but does not fly the ship

Open **Controls...**, check the device and calibration, then verify the axis bindings. A raw joystick needs at least two axes and one button.

MIDI is silent

Check the selected music backend. If using SoundFont MIDI, make sure a valid `.sf2` file is selected.

The game closes or crashes while loading

Start it once with `--gpu-debug` (or set the environment variable `OPENPRIV_GPU_DEBUG=1`):

```
openprivateer --gpu-debug
```

The graphics device then checks every graphics call and writes the graphics driver's own error messages to the session log in the `logs` folder. Loading and play are slower in this mode, so use it only to investigate. Send that log with your report; its first lines also name the graphics driver in use (for example `GPU device driver: vulkan`).

When the game has to stop because of an error, it shows a message with the error and the path of the session log before it closes.

Missing pictures or graphics errors on Linux and macOS

The NVIDIA graphics driver on Linux keeps one open file per piece of graphics memory, and the game's pictures alone come close to the usual default limit of 1024 open files. So on Linux and macOS the game raises its own open-file limit at every start, as far as the system allows. The first lines of the session log show the result:

```
open file limit: soft 1024 hard 1048576 -> raised to 1048576
```

`already at` means the limit was high enough already. `could not raise` means the system refused; then raise the limit before starting the game, for example with `ulimit -n 65536` in the terminal you start it from (`ulimit -Hn` shows the highest value allowed), or with a `nofile` entry in `/etc/security/limits.conf`. On Linux the log also gives the number of open files at several points (`open file descriptors ...`).

A picture the graphics card cannot take is left blank and the game goes on; the log then has a warning with `SDL_CreateGPUTexture failed` or `SDL_CreateGPUTransferBuffer failed`. Send that log with your report.

Something behaves differently from DOS Privateer

Check the Experience mode first. Use **Classic** for direct comparisons with the original.

If the difference remains in Classic, report it.

Windows Defender reports the executable

New unsigned software can occasionally trigger heuristic antivirus detections. Do not blindly whitelist a file.

Make sure the binary came from the official OpenPrivateer release, then update Windows Security's security intelligence and scan it again. A signature update can clear such a detection.

If a current official build is still detected, report the build ID and exact detection name.

macOS says the program cannot be opened

The macOS packages are not signed by Apple, so macOS blocks the first start of a downloaded copy with a warning. Make sure the archive came from the official downloads page, then either:

- try to start `openprivateer` once, then open the security section of macOS's system settings and allow the blocked program there (the names of the settings, the section and the button differ between macOS versions); do the same for the `libSDL3` library if macOS names it next, or
- in Terminal, clear the download mark from the whole program folder:

  ```
  xattr -dr com.apple.quarantine /path/to/openprivateer-folder
  ```

macOS remembers the choice for that copy.

28. Reporting Bugs

A useful Alpha bug report includes:

- OpenPrivateer build ID
- Privateer or Righteous Fire
- Classic, Enhanced or Modern
- what you expected
- what happened
- exact reproduction steps
- whether it happens every time
- the savegame when relevant
- a screenshot or video for visual problems
- for startup or file problems, the text from the launcher's **Copy Diagnostics** button

Print the build ID with:

```
openprivateer --version
```

For base-screen problems, a screenshot with the `R` region overlay can help.

You do not need to decide whether something is an OpenPrivateer bug or an original Privateer quirk before reporting it. Describe what happened and it can be checked against the original.

29. What Alpha Means

OpenPrivateer is playable, but it is still an Alpha.

The normal single-player Privateer path is implemented, and Righteous Fire support is under active reconstruction and testing. Obscure cases are still being found as more people play the game.

Expect bugs, unfinished polish and occasional behavior changes as the original game becomes better understood.

Save before doing anything adventurous.

30. Help and Downloads

Project website:

https://openprivateer.org/

Downloads:

https://openprivateer.org/#downloads

Discord:

https://discord.gg/5H3vW2STqW

When asking for help, include the build ID and whether you are playing Privateer or Righteous Fire.

31. Legal Note

OpenPrivateer is a reimplementation of the game engine.

It does not ship the copyrighted game data from *Wing Commander: Privateer* or *Righteous Fire*. You must provide the original game files yourself.

Wing Commander, *Privateer*, *Righteous Fire* and related names and assets belong to their respective rights holders.

32. One-Minute Checklist

For Privateer:

```
[ ] Extract OpenPrivateer

[ ] Start OpenPrivateer

[ ] Launcher: copy in PRIV.TRE and INITCFG.PAK

[ ] Leave Experience on Classic for original behavior

[ ] Click PLAY, then NEW
```

For Righteous Fire:

```
[ ] Launcher: copy in RF.TRE and INITRF.PAK

[ ] Select Righteous Fire
```

For an old pilot:

```
[ ] Launcher: Files and Diagnostics, Savegames

[ ] Copy .SAV or .PRS there

[ ] Click LOAD
```

For OpenPrivateer's extra features:

```
[ ] Switch to Enhanced or Modern

[ ] Open the settings overlay with ~

[ ] Enable the features you want
```

Have fun.